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Data Communications

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DATA COMMUNICATIONS

Course Information

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Course Objectives

- Introduce the concept of data communication and Internet.
- Familiarize with the features of Data Link Layer.
- Explore the protocols of Internetwork and Routing.
- Understand the working of Transport and Application Layer Protocols

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Course Outcomes



- Comprehend the usage of Layered architecture in Data Communication.
- Implement different error detection schemes and MAC Protocols.
- Comprehend the usage of Internetworking and Routing protocols.
- Decide on the Transport and Application Layer Protocols for communication.

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Course Overview



- This course is an introduction to the basic concepts on data communication and the design of computer networks.
- This course starts with discussing the importance of layered architecture and data transmission methodologies.
- It helps in understanding how analog and digital data can be transmitted using either analog or digital signals.
- It deals with the handling of the errors in the transmission and MAC protocols used.
- It gives a clear picture about the reliable communication protocols of the transport layer and also deals with internetworking principles.
- Finally looking into DNS which is an essential name/directory look up service for the internet along with HTTP which is the support protocol on which the World Wide Web (WWW) operates and deals with Multimedia communication on the internet.

Unit 1: Introduction to Data Communication and Data Transmission

- Data communications and Networking for today's enterprise, Communication Model, Data flow: Types of connections, Types of communication methods.
- Protocol Architecture: The need for a protocol architecture, The TCP/IP protocol architecture: functions of each layer, The OSI model.
- Data and signals: Digital signals: Bit rate, bit length, Transmission impairments: Attenuation, distortion, noise. Digital-to-digital conversion: Unipolar, Polar -NRZ L, NRZ I, RZ, Biphase - Manchester, Differential Manchester encoding. Analog-to-digital conversion: pulse amplitude and pulse code modulation.
- **Experiential Learning:** Problem Solving sessions on digital-to-digital, Analog-to-digital conversion methods (worksheets)

Unit 2: Bandwidth Utilization and Data link layer

- Introduction to Bandwidth and Throughput - Definitions: Bandwidth, Throughput, Latency, Factors affecting bandwidth and throughput. Bandwidth Utilization Concepts: Efficient use of bandwidth, Bit rate vs. Baud rate
- Multiplexing: Frequency division multiplexing, synchronous time division multiplexing, statistical time division multiplexing.
- Data link layer: introduction, link layer addressing- three types of addresses, Address resolution protocol. Standard Ethernet: characteristics, addressing, access method- CSMA/CD, Wireless LANs – Architecture, Addressing mechanism
- Framing Techniques - Character count - Byte stuffing - Bit stuffing
- **Experiential Learning:** Problem solving sessions with Framing, Bandwidth and throughput

Unit 3: Internetworking

- Internetwork protocols: Principles of internetworking, Devices in Internetworking (Router, Switch, Bridge, Gateway)
- Internet protocol operations: internet protocol, IPv4, CIDR, Virtual Private Networks and IP security, Dynamic Host Configuration Protocol(DHCP).
- Routing protocols – Routing Information Protocol, Open Shortest Path First.
- **Experiential Learning:** Static and dynamic routing using Packet tracer, DHCP

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Course Contents



Unit 4: Transport and Application Layer Protocols

- **Transport protocols: Connection-oriented transport protocol mechanisms - TCP, connectionless Transport-UDP.**
- Internet applications: Internet directory services- DNS, Web access- HTTP – Status codes
- Multimedia Applications: Audio and video compression, Voice over IP and multimedia support – SIP.
- **Experiential Learning:** Study of HTTP and DNS using Packet tracer

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Text Books and Reference Books



Text Books:

1. B. A. Forouzan, Data Communications and Networking with TCP/IP Protocol Suite, 6th Ed. New York, NY, USA: McGraw-Hill Education, 2021.

Reference Book:

1. A. S. Tanenbaum, N. Feamster, and D. J. Wetherall, Computer Networks, 6th Ed. Boston, MA, USA: Pearson Education, 2021.
2. W. Stallings, Data and Computer Communications, 10th Ed. Boston, MA, USA: Pearson Education



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Introduction

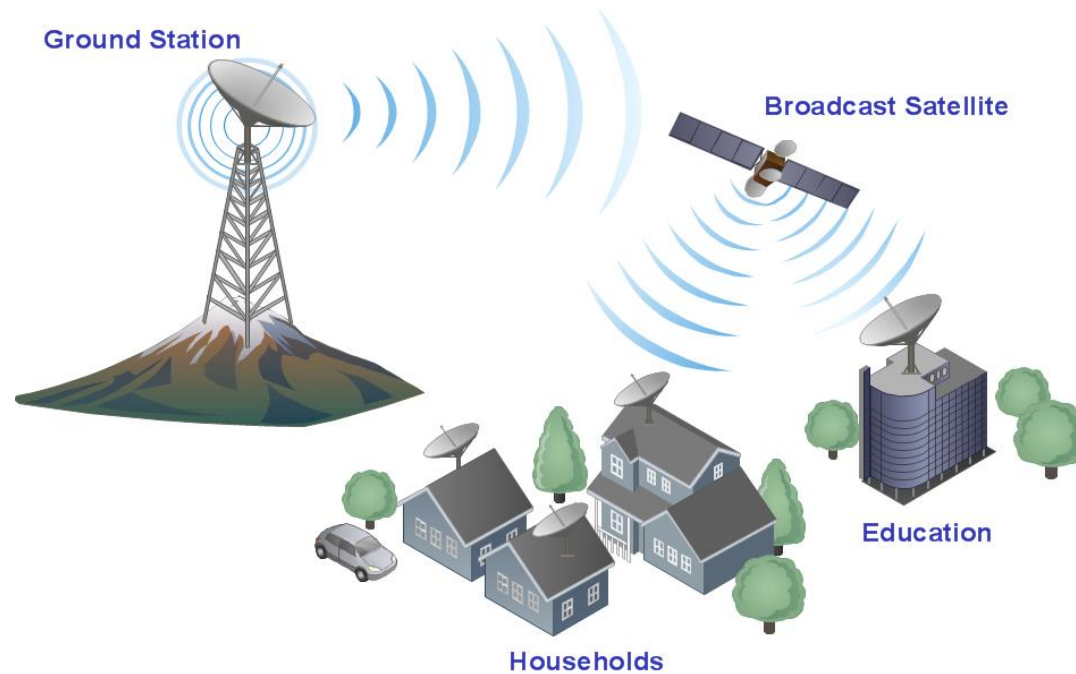
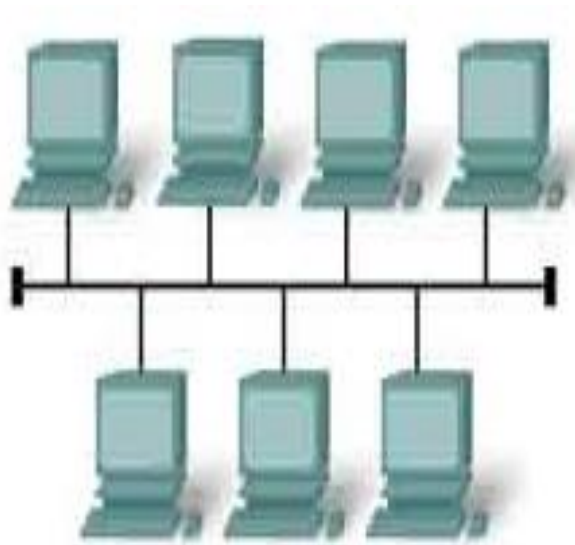
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Data Communication

- Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable or Wireless.



Data Communications

Data Communications and Networking for Today's Enterprises



Forces behind the evolution of data communications and networking :

- Data Communications & Networking for Today's Enterprise revolves around enabling efficient, reliable, and secure communication in a digital business environment.
- Data communication and networking are the backbone of digital enterprises, enabling **speed, reliability, security, and global connectivity** essential for business success.
- Modern enterprises rely heavily on **data communication and networking** to operate efficiently, connect teams, manage resources, and deliver services globally.
- Today's enterprises are strategic enablers of Innovation, Agility, Competitiveness

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Data Communications and Networking for Today's Enterprises



Traffic growth at a high and steady rate:

- Communication traffic – local (within a building or building complex) and long distance, both voice and data.
- The increasing office automation, remote access, online transactions etc

Development of new services

- Demand for high-capacity networking and transmission facilities

Advances in Technology

- The trend towards faster and cheaper both in computing and communications. Eg: Mbps, Gbps, Tbps
- Network offers different levels of QoS – throughput, delay, security
- Innovations includes email, remote data access, high speed wireless access.

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Enterprise Applications

- **Unified Communication (UC):** Integration of voice, video, and messaging
- **CRM/ERP Integration:** Seamless data exchange across platforms
- **Remote Access:** VPNs, remote desktops, and cloud access
- **Security Systems:** Firewalls, intrusion detection/prevention systems (IDS/IPS)

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Data Communications and Networking for Today's Enterprises



Technologies Used

- **Ethernet:** Primary wired LAN technology (IEEE 802.3)
- **Wi-Fi:** Wireless connectivity standard (IEEE 802.11)
- **Cloud Networking:** Offloading infrastructure to services like AWS, Azure
- **SDN (Software-Defined Networking):** Centralized network control
- **IoT (Internet of Things):** Smart device interconnectivity
- **5G:** High-speed mobile communication for real-time data

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Data Communications and Networking for Today's Enterprises



Key Components

- **Networks:** LAN, WAN, MAN, WLAN, and VPN
- **Protocols:** TCP/IP, HTTP/HTTPS, FTP, SMTP, DHCP, DNS
- **Hardware:** Routers, Switches, Hubs, Access Points, Firewalls
- **Transmission Media:** Fiber optics, twisted pair cables, wireless

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Communication Model

Five components of Data Communication

- **Sender:** The device sends a message.

Example: computer, smartphone, sensor, camera

- **Receiver:** The **device** that receives the message.

Example: computer, mobile phone, printer, etc.

- **Message:** Information(data) to be communicated

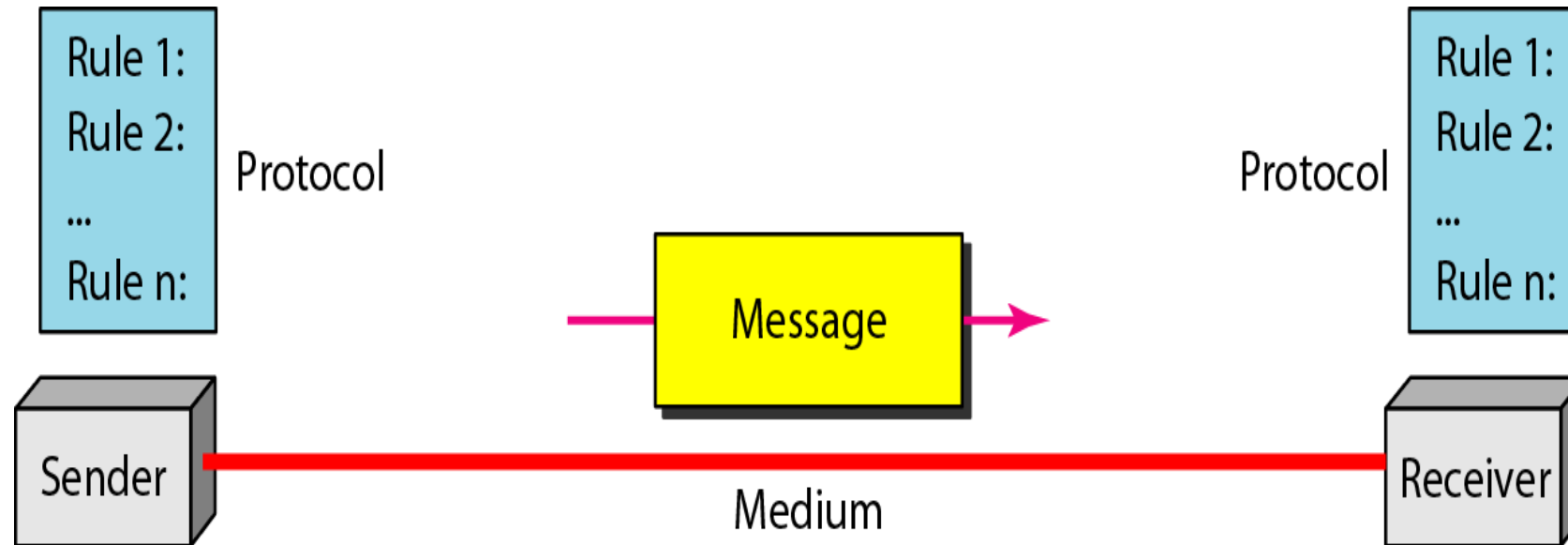
- **Transmission medium:** Physical path by which a message travels

Examples: Twisted-pair wire, Coaxial cable, Fiber optics, Radio waves

- **Protocol:** A set of rules that govern data communication Data Communications.

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Communication Model



- *Text*
- *Numbers*
- *Images*
- *Audio*
- *Video*

Text

- In data communications, text is represented as a bit pattern, a sequence of bits.
- Text is represented as a sequence of bits(Os or Is), using coding systems like ASCII and Unicode.

Numbers

- Numbers are also represented by bit patterns.
- However, a code such as ASCII is not used to represent numbers; the number is directly converted to a binary number to simplify

Images

- Images are also represented by bit patterns. In its simplest form, an image is composed
- of a matrix of pixels (picture elements), where each pixel is a small dot.
- The size of the pixel depends on the *resolution*.
- There are several methods to represent color images(RGB, YCM)

Audio

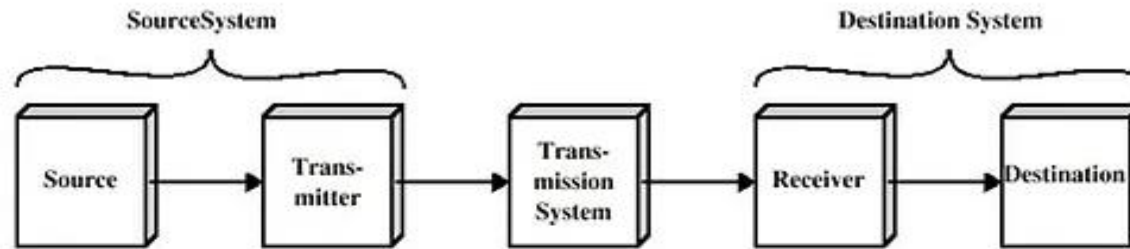
- Audio refers to the recording or broadcasting of sound or music.
- Audio is by nature different from text, numbers, or images.
- It is continuous, not discrete.
- Even when we use a microphone to change voice or music to an electric signal, we create a continuous Signal

Video

- Video refers to the recording or broadcasting of a picture or movie.
- Video can either be produced as a continuous entity (e.g., by a TV camera), or it can be a combination of images, each a discrete entity, arranged to convey the idea of motion.
- Again we can change video to a digital or an analog signal.

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Communication Model



(a) General block diagram



(b) Example

Data Communications

Communication Model

- **Source:** generates the data to be transmitted
- **Transmitter:** encodes the information
- **Transmission System:** single transmission line or complex network connecting source and destination
- **Receiver:** Accepts the signal from transmission system and converts it into a form that can be handled by destination device.
- **Destination:** Takes the incoming data from the receiver



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Types of Connections & Communication Methods

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Data flow: Types of connections

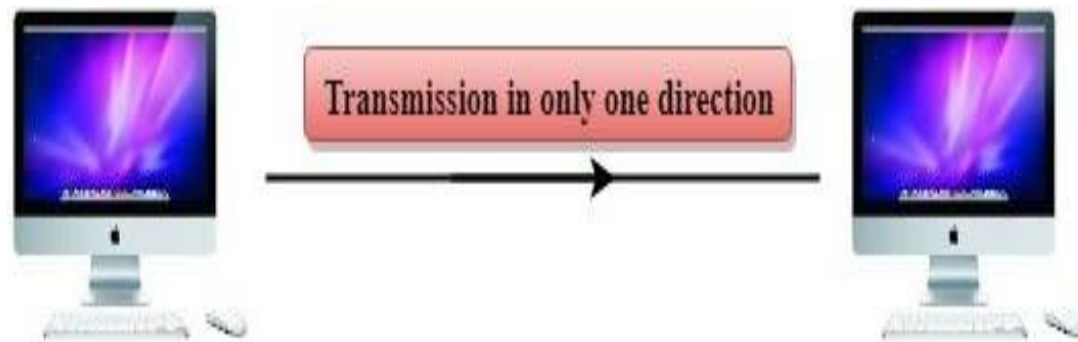
- **Simplex Mode**
- **Half Duplex Mode**
- **Full Duplex Mode**

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Data flow: Types of connections

Simplex Mode

- The communication is unidirectional
- Only one of the two devices on a link can transmit; the other can only receive.
- Example: Keyboards and traditional monitors

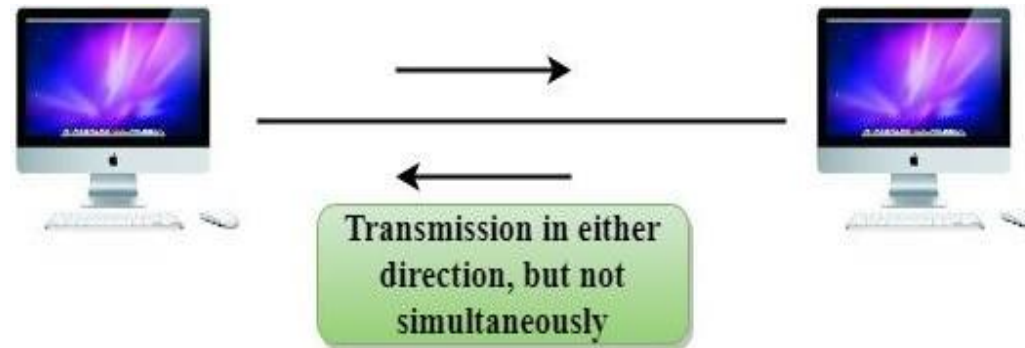


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Data flow: Types of connections

Half-duplex mode

- Each station can transmit and receive, but not at the same time.
- When one device is sending, the other can only receive, and vice versa
- Eg: Walkie-talkies

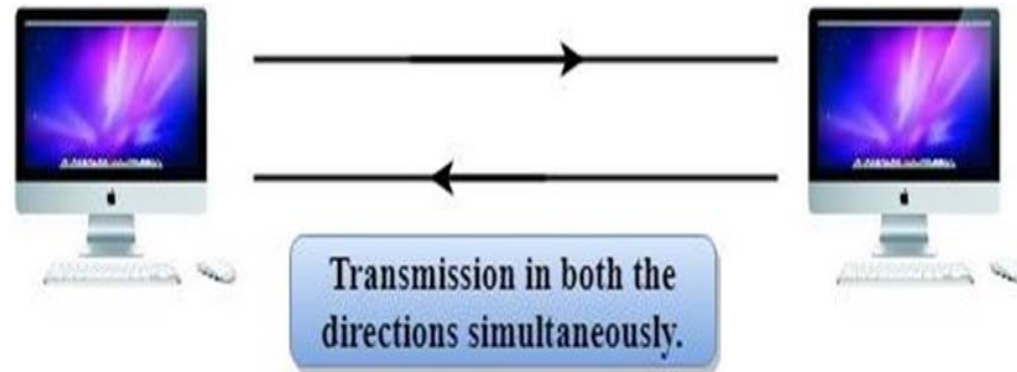


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Data flow: Types of connections

Full duplex mode

- Both stations can transmit and receive simultaneously
- Example : Telephone network
- When two people are communicating by a telephone line, both can talk and listen at the same time.



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Data flow: Types of connections



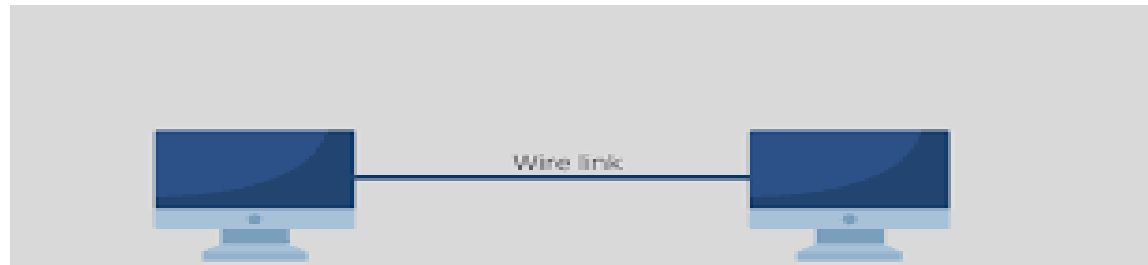
- A network is two or more devices connected through links. A link is a communications pathway that transfers data from one device to another.
- **Point to point**
- **Multipoint**

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Data flow: Types of connections

Types of connections – Point to point

- A point-to-point connection provides a dedicated link between two devices.
- The entire capacity of the link is reserved for transmission between two devices.
- Point-to-point connections use cable or infrared to connect the two ends.
- Ex: television channels are changed by remote control, it establishes a point-to-point connection between the remote control and the television's control system.

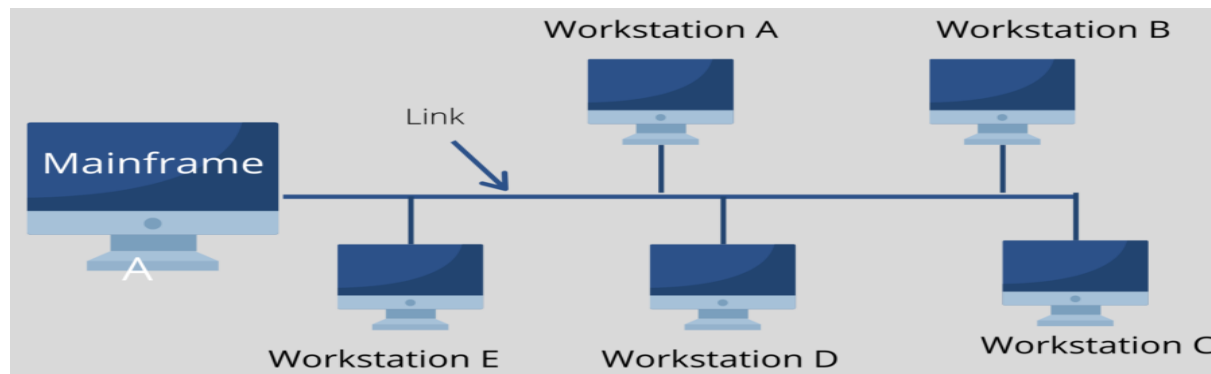


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Types of communication methods

Types of connections - Multipoint

- Also called **multidrop**
- more than two specific devices share a single link.
- The capacity of the channel is shared, either spatially or temporally.
- If several devices can use the link simultaneously, it is a spatially shared connection. If users must take turns, it is a timeshared connection



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Types of communication methods

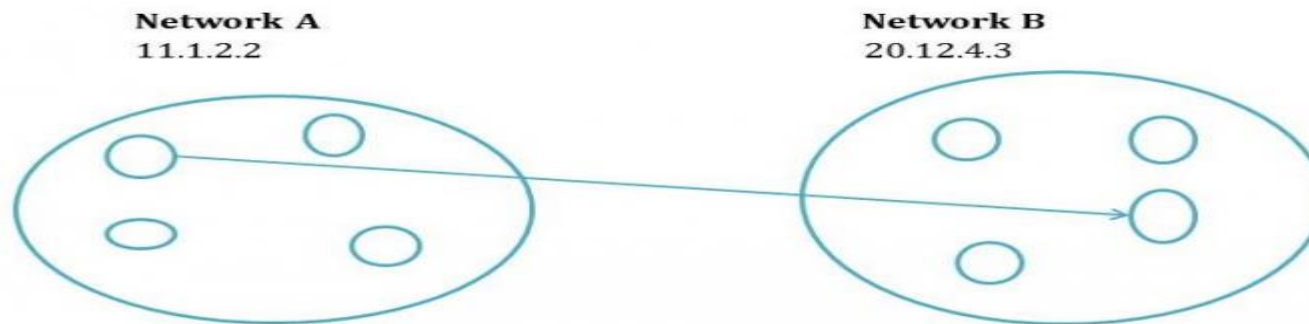
- There are multiple ways to transmit data between the connected nodes.
- Transmission can be one-to-one, one-to-many, or many-to-many. This transmission of data in the network is called casting. Casting is of 3 types:
 - **Unicasting**
 - **Broadcasting**
 - **Multicasting**

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Types of communication methods

Unicasting

- In Unicast transmission, the data is transferred from a single sender (or a single source host) to a single receiver (or a single destination host)
- Unicast transmission is a type of one-to-one communication.
- The sender node is from one network whereas the receiver node belongs to another



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Types of communication methods

Limited Broadcasting

- The data is transmitted from a single source host to all the other hosts residing in the same network.

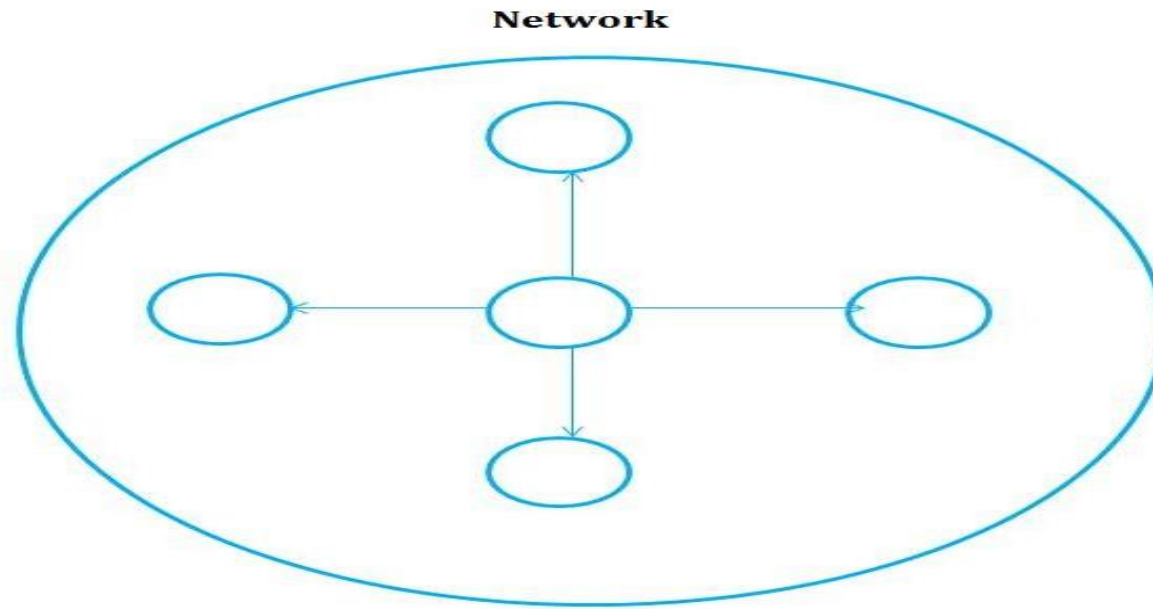


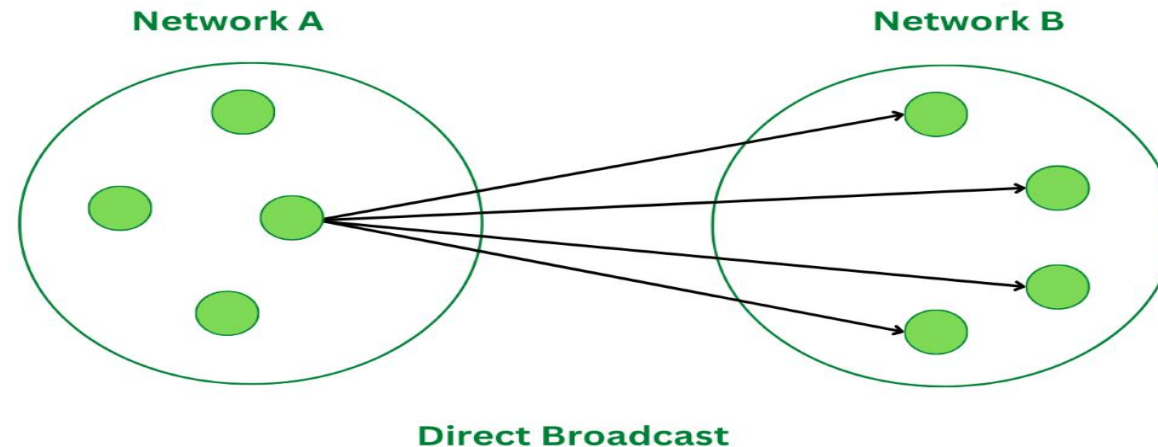
Figure: Limited Broadcast

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Types of communication methods

Directed Broadcasting

- Directed Broadcast transmits data from one source host to all the other hosts that exist in some other network.
- This is used when all the receiving hosts from another network require the same data

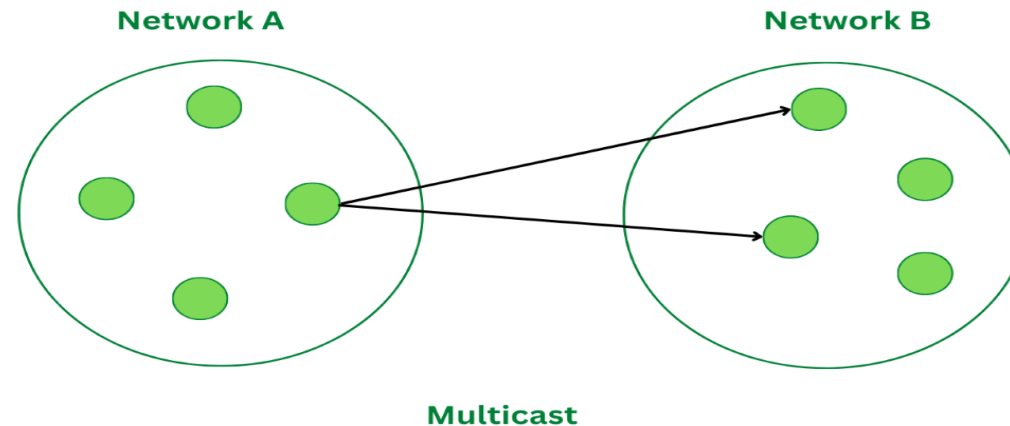


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Types of communication methods

Multicasting

- In multicasting one or more sending hosts sends the data packets to one or more receiving hosts in another network.
- This type of transmission is known as one-to-many communication.



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Network Topology



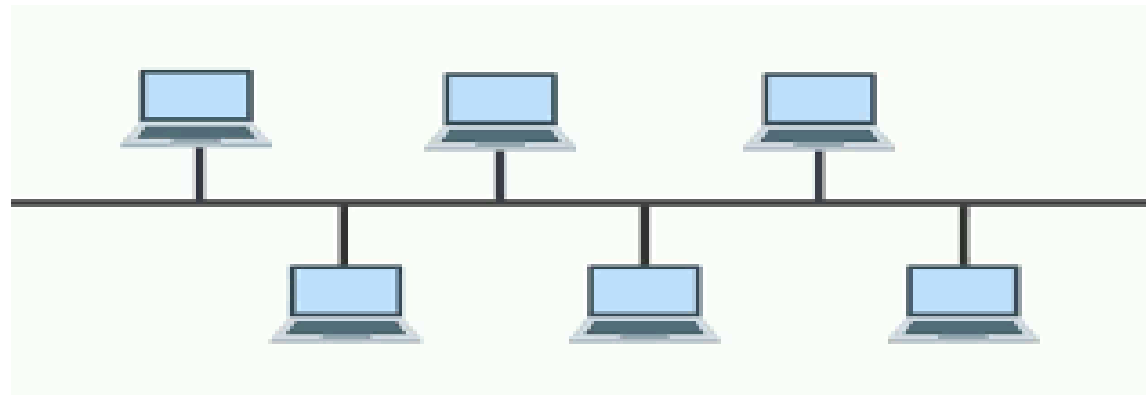
A network topology is defined as the layout or arrangement of nodes in a communication network.

Types of topologies:

- Bus topology
- Ring topology
- Star topology
- Tree topology
- Mesh topology
- Hybrid topology

Network topology – BUS

- All the nodes are connected to the single cable.
- The central cable is the backbone of the network and is known as bus.
- A signal from the sender or source is transmitted and it travels to all nodes connected to bus cable.
- Only the intended recipient whose destination address matches accepts it.



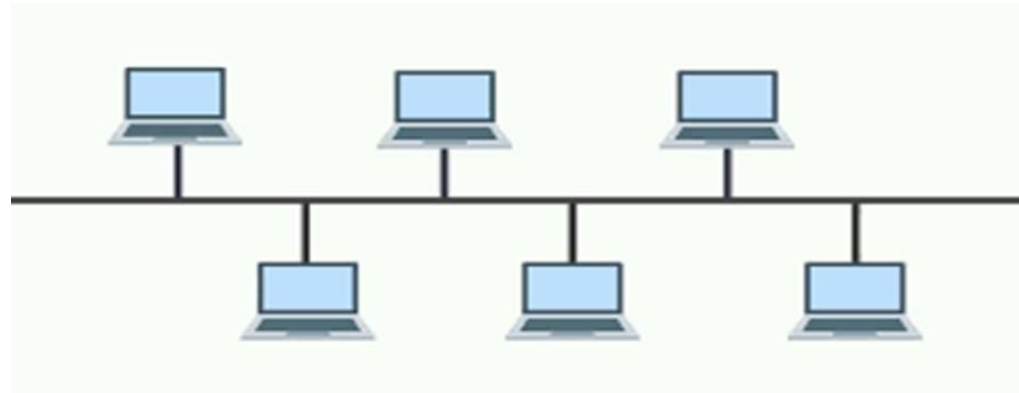
Network topology -BUS

Advantages of Bus Topology

- Good for small network
- Less cable needed to connect the required nodes
- Cost Effective
- Easy to Understand

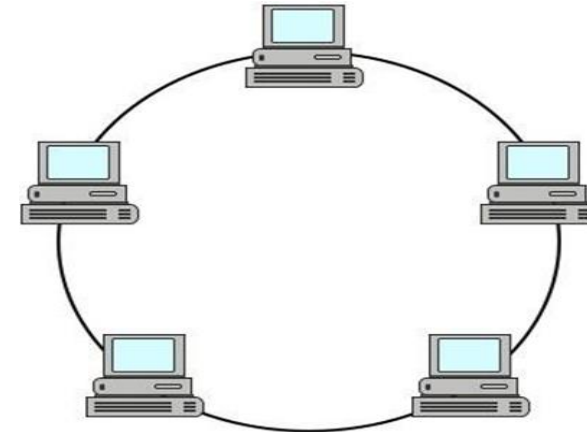
Disadvantages of Bus Topology

- Unidirectional
- If the primary cable fails, the entire network fails



Network topology – RING

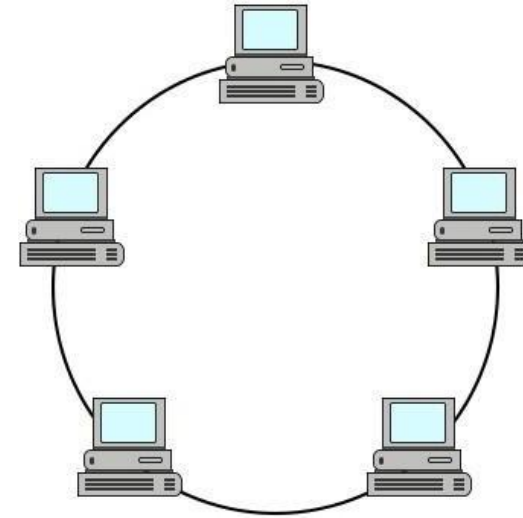
- All nodes are connected to each other in a closed loop.
- A ring can be unidirectional or bidirectional .
- A signal from the source is transmitted to its adjacent node and there the node check its address with the message destination address if the match found the message is accepted otherwise discards it.



Network topology - Ring

Advantages of Ring Topology

- Inexpensive to implement
- Better performance and works with high loads
- Orderly network flow
- Makes it easier to detect faults

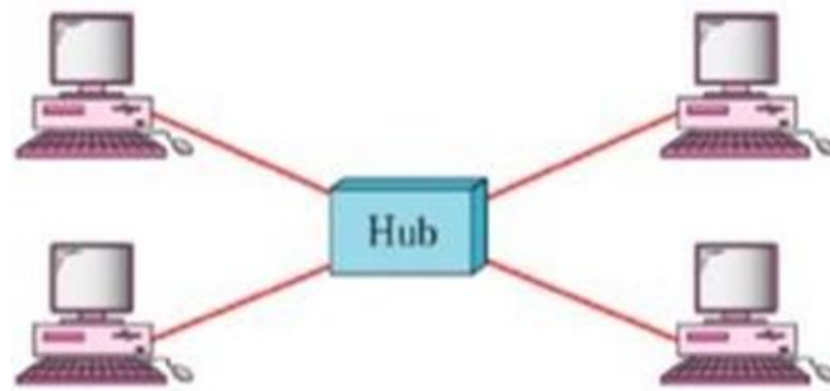


Disadvantages of Ring Topology

- Adding, removing, or configuring nodes requires the network to be shut down
- One faulty node can destroy the entire network

Network topology – STAR

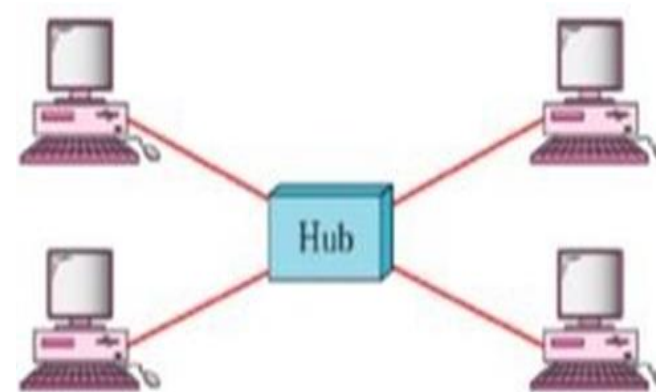
- All the nodes of networks are connected to the central device called hub.
- workstations are connected to hub with a point to point connection.
- All the data on star topology passes through the central device before reaching the intended destination. Hub acts as a junction to connect different nodes present in star network.
- Star network is widely used in banks.



Network topology – STAR

Advantages of Star Topology

- Easy to install and modify
- Devices can be added, removed, or modified without slowing down the central hub.
- Less cabling needed and easy to detect faults

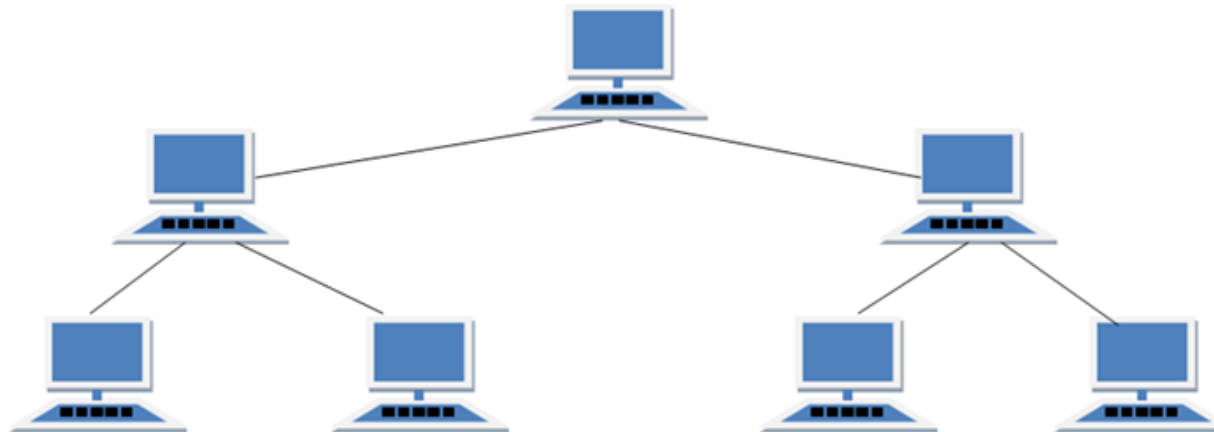


Disadvantages

- If the central hub fails, all of the connected nodes are disabled
- Performance is based on the hub's efficiency, power, and configuration

Network topology – TREE

- Tree topologies have a root node, and all other nodes are connected which form a hierarchy.
- Also called as hierarchical topology.
- Tree topology integrates the characteristics of star and bus topology.



Network topology – TREE

Advantages:

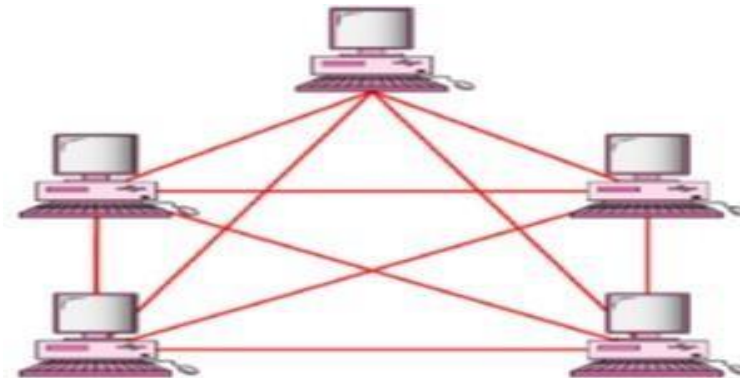
- Failure of one node never affects the rest of the network.
- Node expansion is fast and easy.
- Detection of error is an easy process

Disadvantages:

- It is heavily cabled topology
- If more nodes are added, then its maintenance is difficult
- If the hub or concentrator fails, attached nodes are also disabled.

Network topology – MESH

- Here each of the network node, computer and other devices are interconnected with one another.
- Communication between nodes is possible either directly or by passing through.
- This topology is ideal for long distances networking because it provides extensive backup and rerouting capabilities.



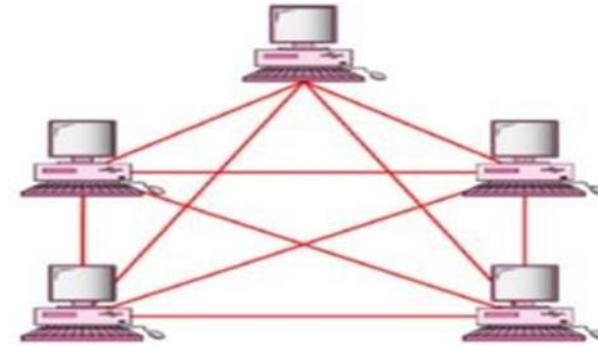
Network topology – MESH

Advantages of Mesh Topology

- High level of redundancy (robust)
- Provides privacy and security
- Fault detection and isolation are easy

Disadvantages of Mesh Topology

- Cost of cabling and installment are high
- Reconfiguring or installation is a long task



Network topology – HYBRID

- Hybrid topology combines two or more topologies.
- A hybrid topology is always produced when two different basic network topologies are connected.





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Protocol Architecture

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The need for a protocol architecture

- The main aim of the layered architecture is to divide the design into small pieces.
- Each lower layer adds its services to the higher layer to run the applications.
- It provides modularity and clear interfaces, i.e., provides interaction between subsystems.
- It ensures the independence between layers by providing the services from lower to higher layer without defining how the services are implemented. Thus any modification in a layer will not affect the other layers.
- The number of layers, functions, contents of each layer will vary from network to network.

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Basic Elements of Layered Architecture

- **Service:** It is a set of actions that a layer provides to the higher layer.
- **Protocol:** It defines a set of rules that a layer uses to exchange the information with peer entity.
- **Interface:** It is a way through which the message is transferred from one layer to another layer.

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Protocol

- *A protocol is synonymous with rule. It consists of a set of rules that govern data communications. It determines what is communicated, how it is communicated and when it is communicated.*

The key elements of a protocol are syntax, semantics and timing

- **Syntax**
 - **Structure or format of the data**
 - **Indicates how to read the bits - field delineation**
- **Semantics**
 - **Interprets the meaning of the bits**
 - **Knows which fields define what action**
- **Timing**
 - **When data should be sent and what**
 - **Speed at which data should be sent or speed at which it is being received.**

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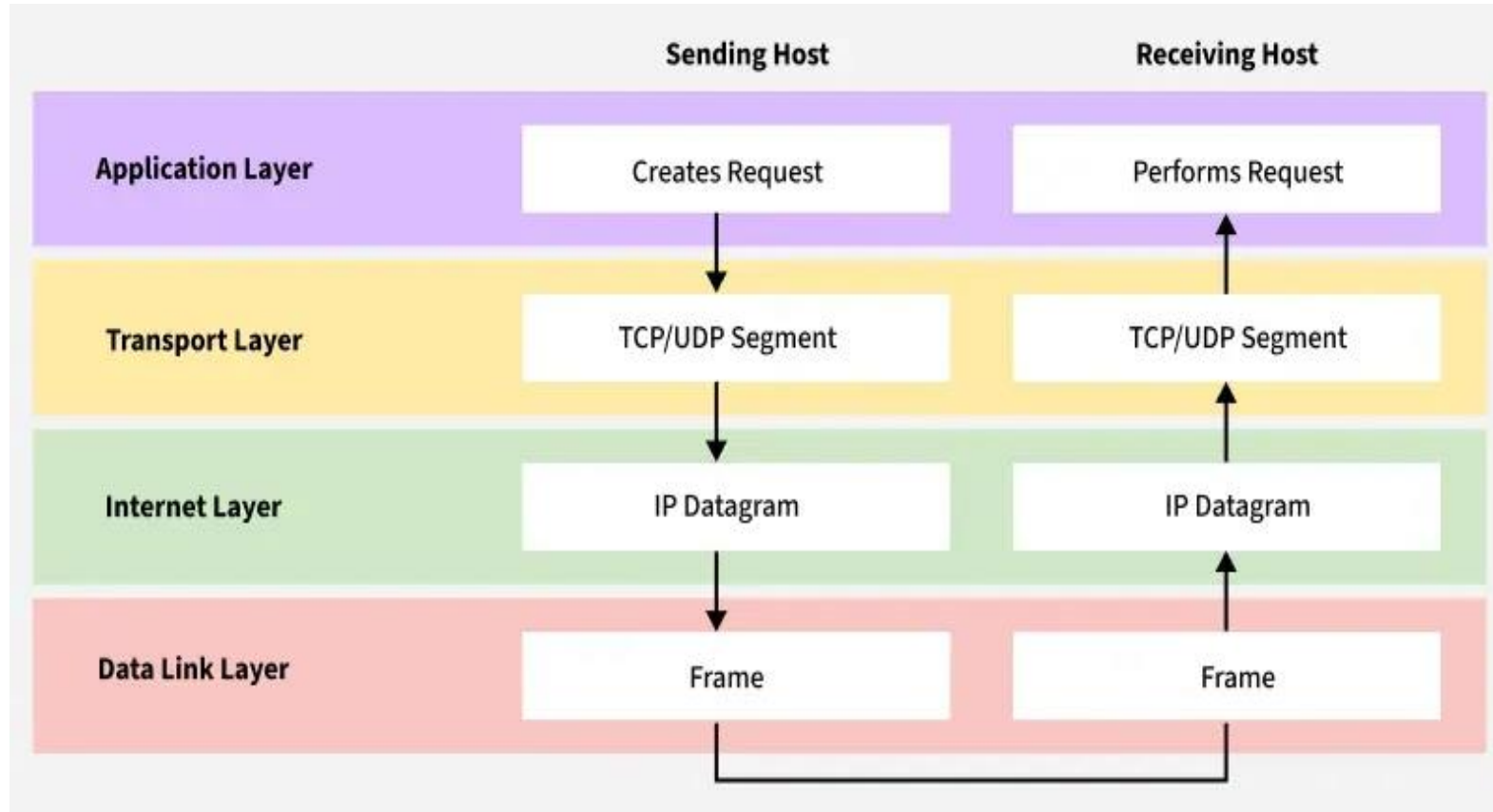
TCP/IP Protocol Architecture

The TCP/IP model consists of five layers:

- Application layer
- Transport layer
- Network layer
- Data link layer
- Physical layer

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TCP/IP Protocol Architecture



When Sending Data (From Sender to Receiver)

- **Application Layer:** Prepares user data using protocols like HTTP, FTP, or SMTP.
- **Transport Layer (TCP/UDP):** Breaks data into segments and ensures reliable (TCP) or fast (UDP) delivery.
- **Internet Layer (IP):** Adds IP addresses and decides the best route for each packet.
- **Link Layer (Network Access Layer):** Converts packets into frames and sends them over the physical network.

When Receiving Data (At the Destination)

Link Layer: Receives bits from the network and rebuilds frames to pass to the next layer.

Internet Layer: Checks the IP address, removes the IP header, and forwards data to the Transport Layer.

Transport Layer: Reassembles segments, checks for errors, and ensures data is complete.

Application Layer: Delivers the final data to the correct application (e.g., displays a web page in the browser).

Application Layer

- An application layer is the topmost layer in the TCP/IP model.
- It allows the user to interact with the application.

The different protocols of this layer:

- **HTTP:** HTTP stands for Hypertext transfer protocol. It allows to access the data over the www. It transfers the data in the form of plain text, audio, video.
- **SNMP:** SNMP stands for Simple Network Management Protocol. It is a framework used for managing the devices on the internet

Application Layer

- **SMTP:** SMTP stands for Simple mail transfer protocol. It is used to send the data to another e-mail address.
- **DNS:** DNS stands for Domain Name System. It maps the name to the address.
- **TELNET:** Terminal Network. It establishes the connection between the local computer and remote computer in such a way that the local terminal appears to be a terminal at the remote system.
- **FTP:** FTP stands for File Transfer Protocol. FTP is used for transmitting the files from one computer to another computer.

Transport Layer

- The TCP/IP transport layer protocols exchange data receipt acknowledgments and retransmit missing packets to ensure that packets arrive in order and without error.
- End-to-end communication.
- Transmission Control Protocol (TCP) and User Datagram Protocol(UDP)are transport layer protocols.
- TCP: Connection oriented protocol, reliable.
- UDP: Connectionless protocol, un reliable, out of order delivery may take place.

Network Layer or Internet layer

- An internet layer is the second layer of the TCP/IP model.
- An internet layer is also known as the **network layer**.
- The main responsibility of the internet layer is to send the packets from any network, and they arrive at the destination irrespective of the route they take.

The main protocols are:

- **IP: Internet Protocol**, is responsible for delivering packets from the source host to the destination host by looking at the IP addresses in the packet headers. IP has 2 versions: IPv4 and IPv6

Network Layer

ICMP: Internet Control Message Protocol. It is encapsulated within IP datagrams and is responsible for providing hosts with information about network problems.

ARP: ARP stands for Address Resolution Protocol. Its job is to find the hardware address of a host from a known IP address.

ARP has several types: Reverse ARP, Proxy ARP and Inverse ARP.

Data Link Layer & Physical

The Data Link layer in the TCP/IP model is responsible for reliable transmission of data frames between two directly connected nodes on a network.

Framing:

The Data Link layer packages data received from the Network layer (IP layer) into frames, adding header and trailer information for addressing and error detection.

Error Detection:

It includes mechanisms like checksums to identify if data has been corrupted during transmission.

Media Access Control (MAC):

This sublayer manages access to the shared network medium, preventing collisions and ensuring fair usage.

Addressing:

The Data Link layer uses physical addresses (MAC addresses) to identify devices on the local network segment.

Flow Control:

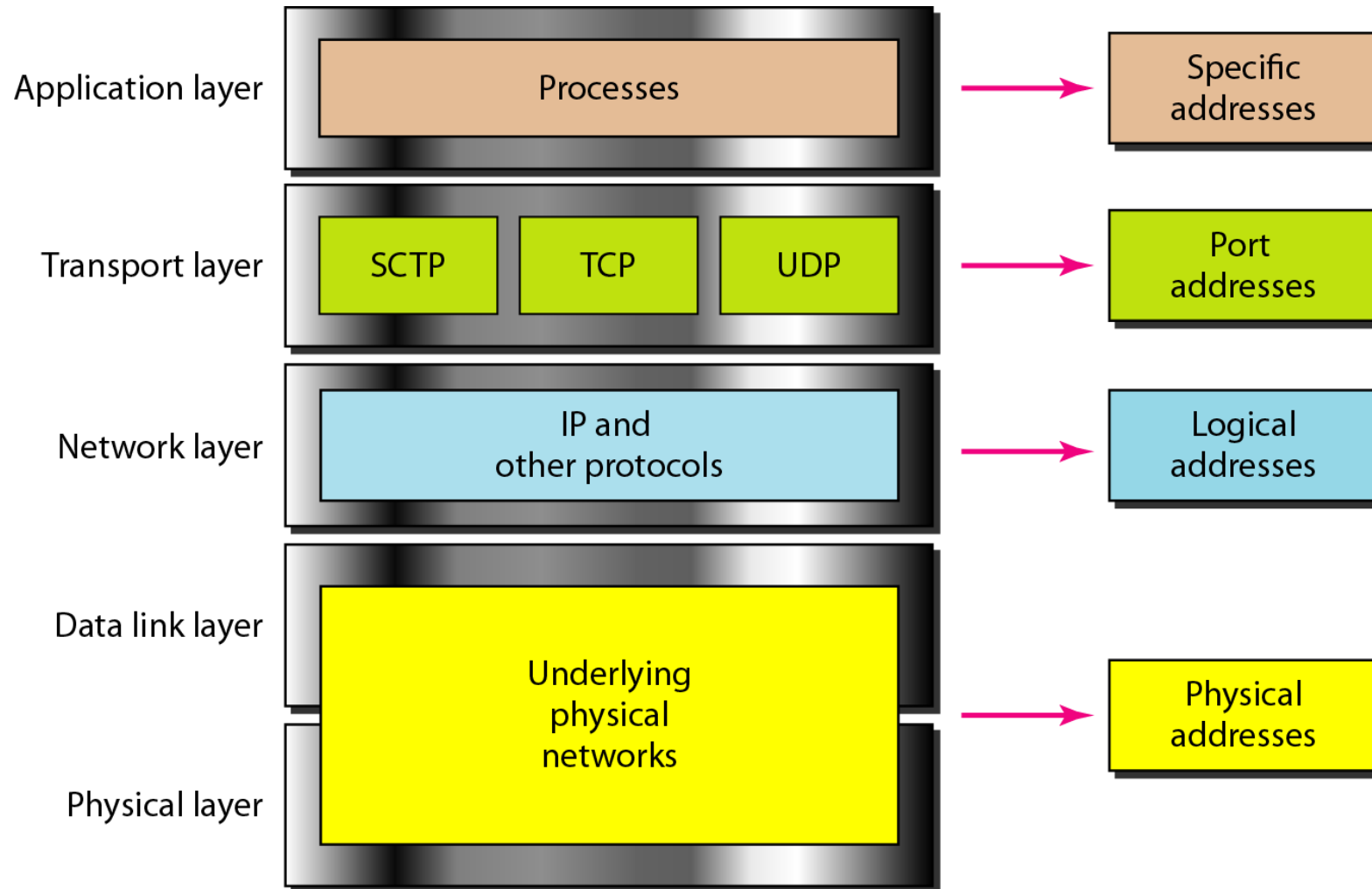
It helps regulate the flow of data between sender and receiver to prevent one from overwhelming the other.

Physical Layer (Layer 1):

- The Physical layer focuses on the physical transmission of raw bits over the network medium, dealing with aspects like electrical signals, cabling, and modulation
- **Bit Transmission:**
- Converts data frames from the Data Link layer into a stream of bits for transmission over the physical medium.
- **Physical Medium:**
- Defines the physical characteristics of the network (cables, wireless signals, etc.) and the way data is represented on it.
- **Signal Encoding:**
- Translates digital data into electrical or optical signals suitable for transmission and back.
- **Synchronization:**
- Ensures proper timing and synchronization between transmitting and receiving devices.
- **Hardware Devices:**
- Includes devices like cables, hubs, modems, and repeaters that facilitate the physical transmission of data

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TCP/IP Protocol Architecture – Protocol & Addresses





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ISO/OSI Protocol Architecture

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OSI & TCP/IP Protocol Architecture – Layers & Protocols



OSI



TCP/IP

SMTP,FTP,
HTTP,POP3,
IMAP4,SNMP

TCP & UDP

IP

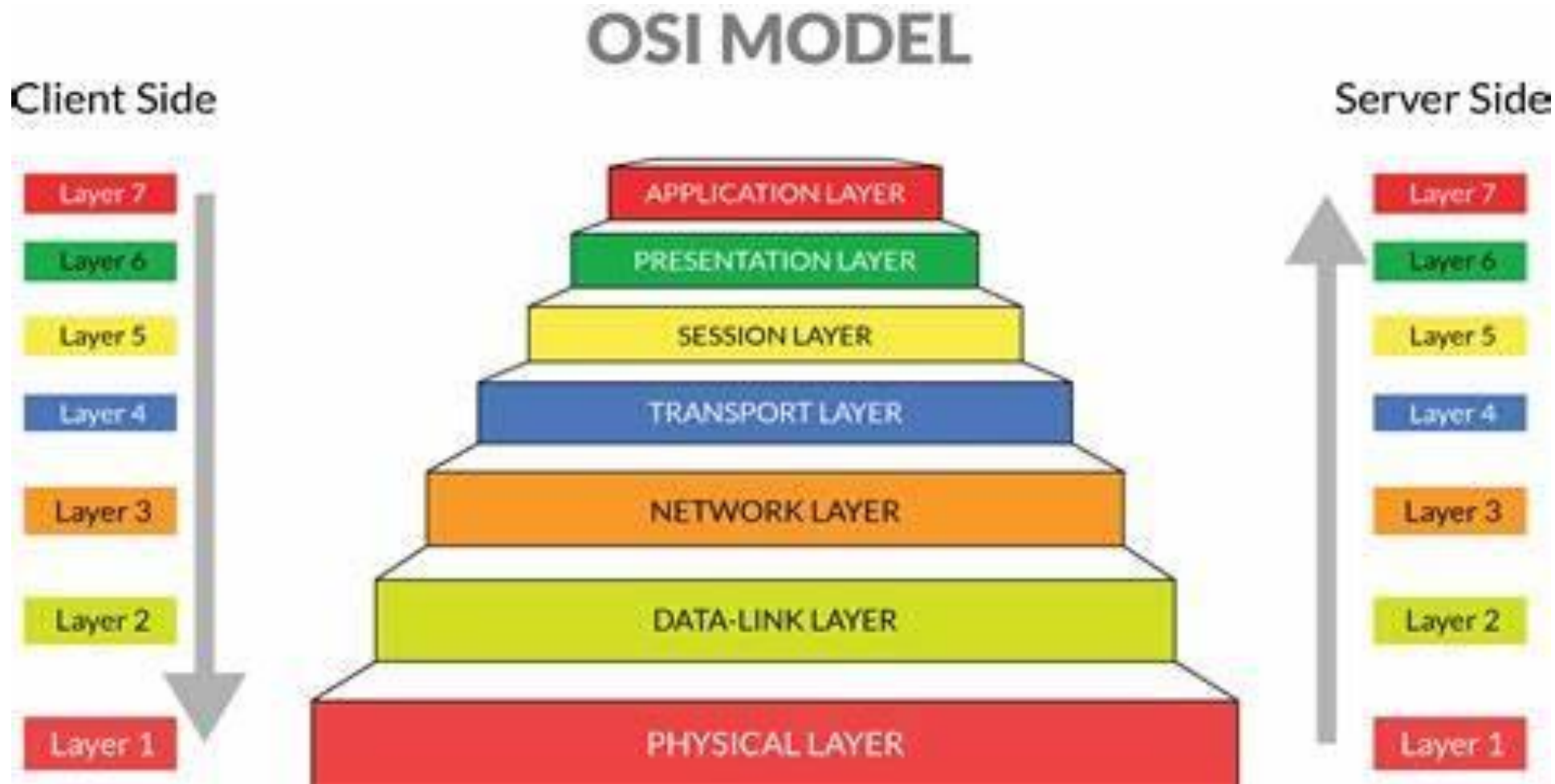
Ethernet

Protocols

- OSI stands for **Open System Interconnection** is a reference model that describes how information from a software application in one computer moves through a physical medium to the software application in another computer.
- OSI model was developed by the International Organization for Standardization (ISO) in 1984
- OSI consists of **seven layers**
- Each layer is self-contained, so that task assigned to each layer can be performed independently.

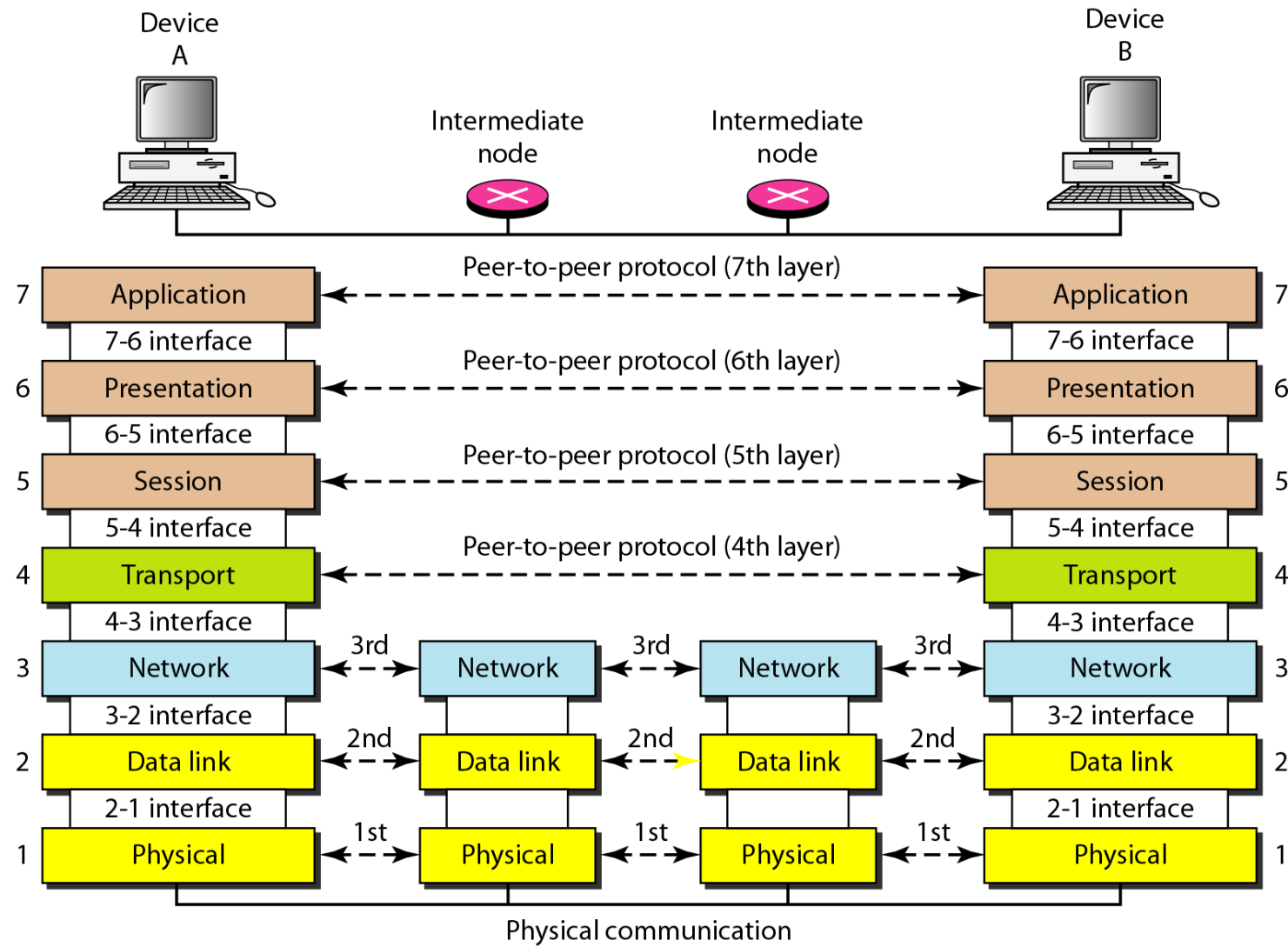
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The OSI Architecture



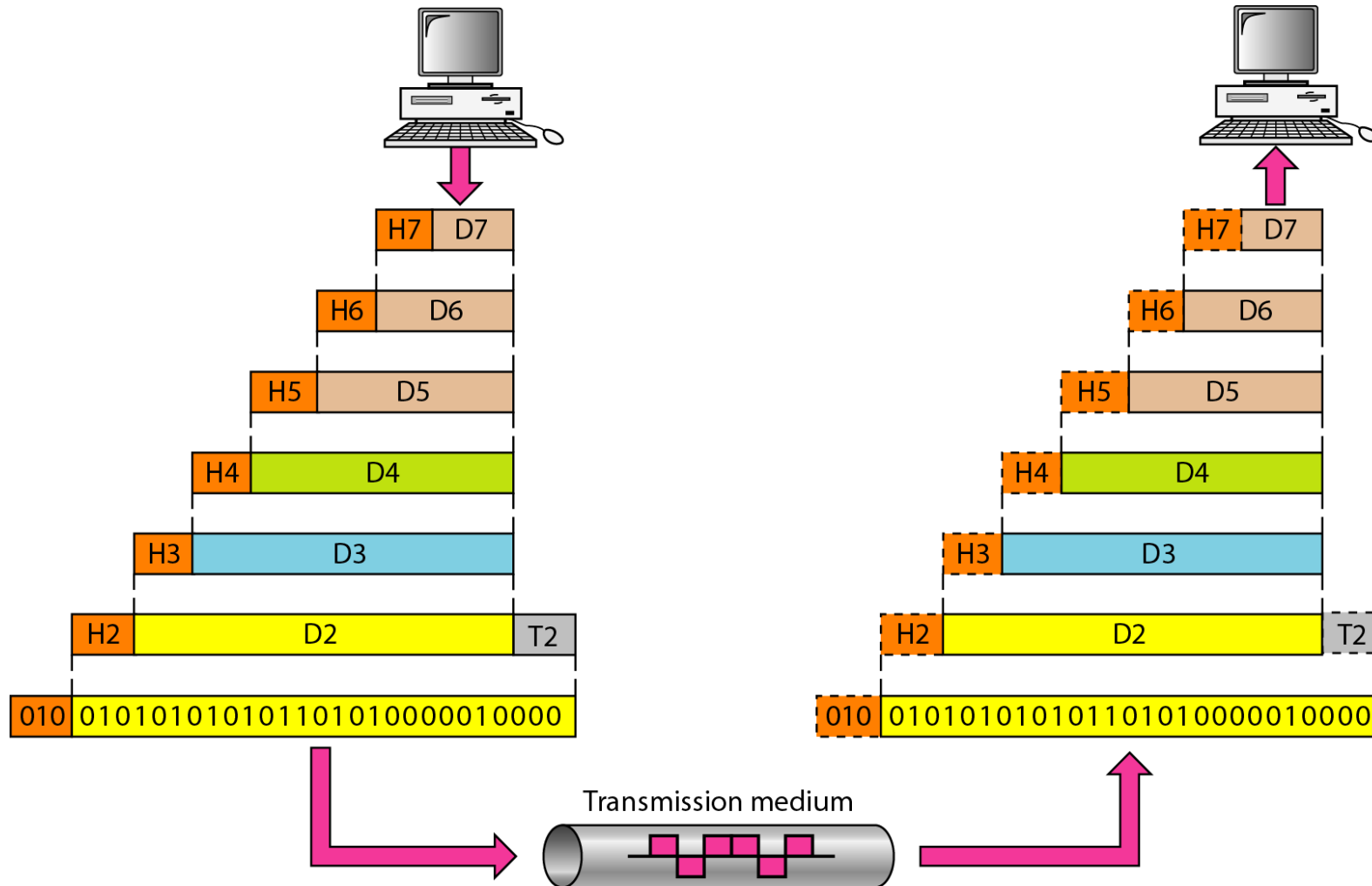
Data Communications

The OSI Architecture



Data Communications

The OSI Architecture



Physical layer:

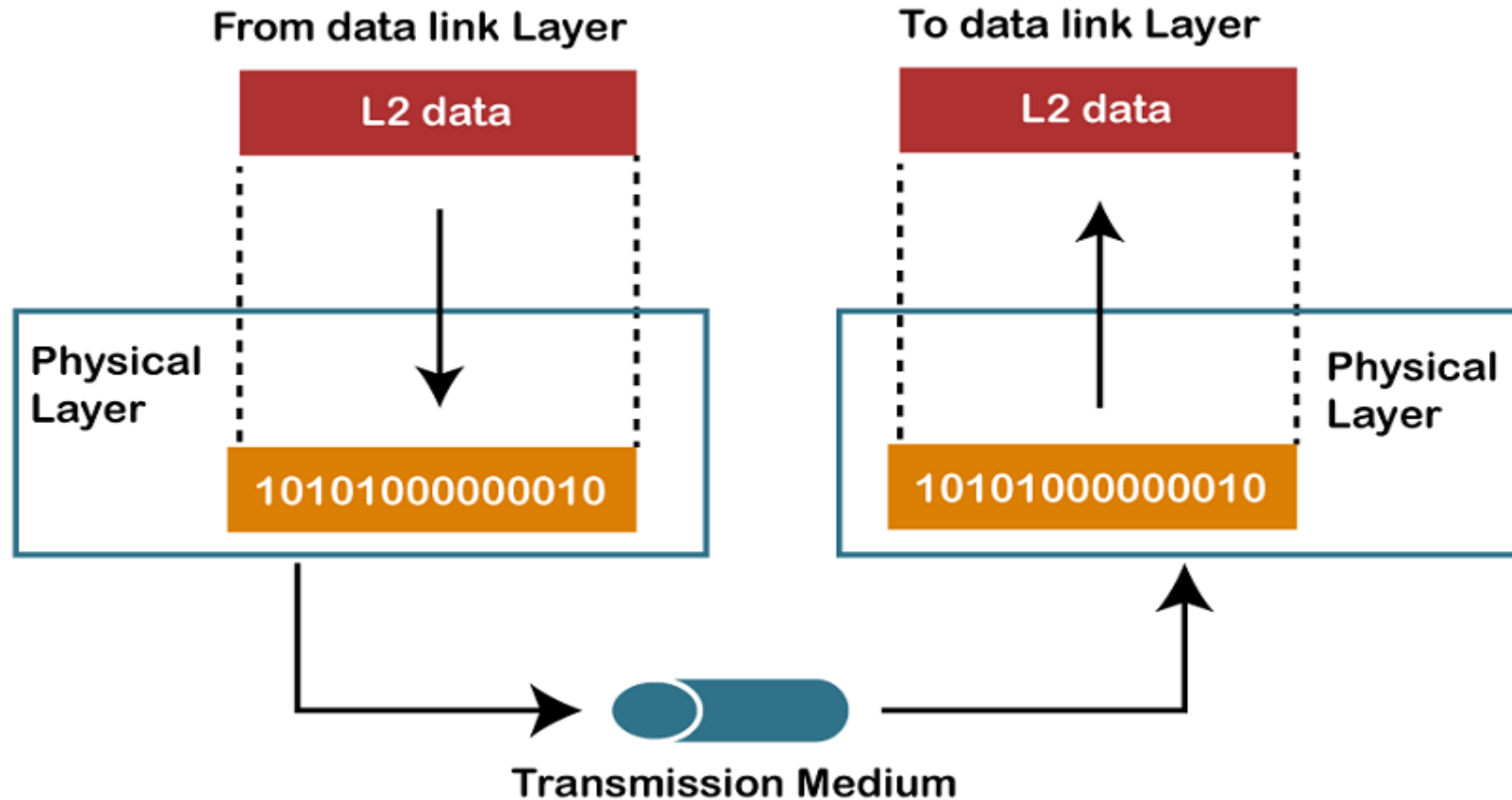
- It is the lowest layer of the OSI model.
- Transmission of individual bits from one node to another node.
- It establishes, maintains and deactivates the physical connection.
- It specifies the mechanical, electrical and procedural network interface specifications.

Functions of a Physical layer:

- **Line Configuration:** defines how two or more devices can be connected physically.
- **Data transmission:** simplex, half-duplex or full-duplex mode
- **Topology:** defines the way how network devices are arranged.
- **Signals:** determines the type of the signal used for transmitting the information.

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The OSI Architecture – Physical layer

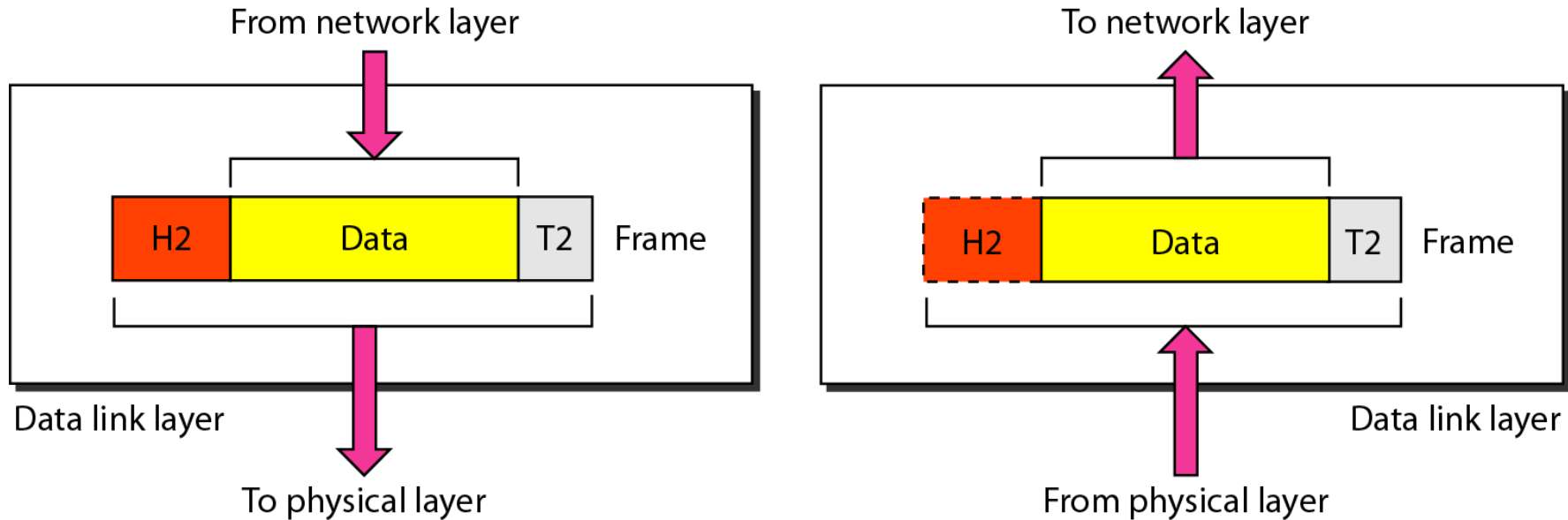


Datalink Layer:

- This layer is responsible for the error-free transfer of data frames.
- It defines the format of the data on the network.
- It provides a reliable and efficient communication between two or more devices.
- It is mainly responsible for the unique identification of each device that resides on a local network.

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The OSI Architecture – Datalink layer



Functions of data link layer

Framing: The data link layer translates the physical's raw bit stream into packets known as Frames. The Data link layer adds the header and trailer to the frame.

Physical Addressing: The frame is transmitted to the destination address mentioned in the header.

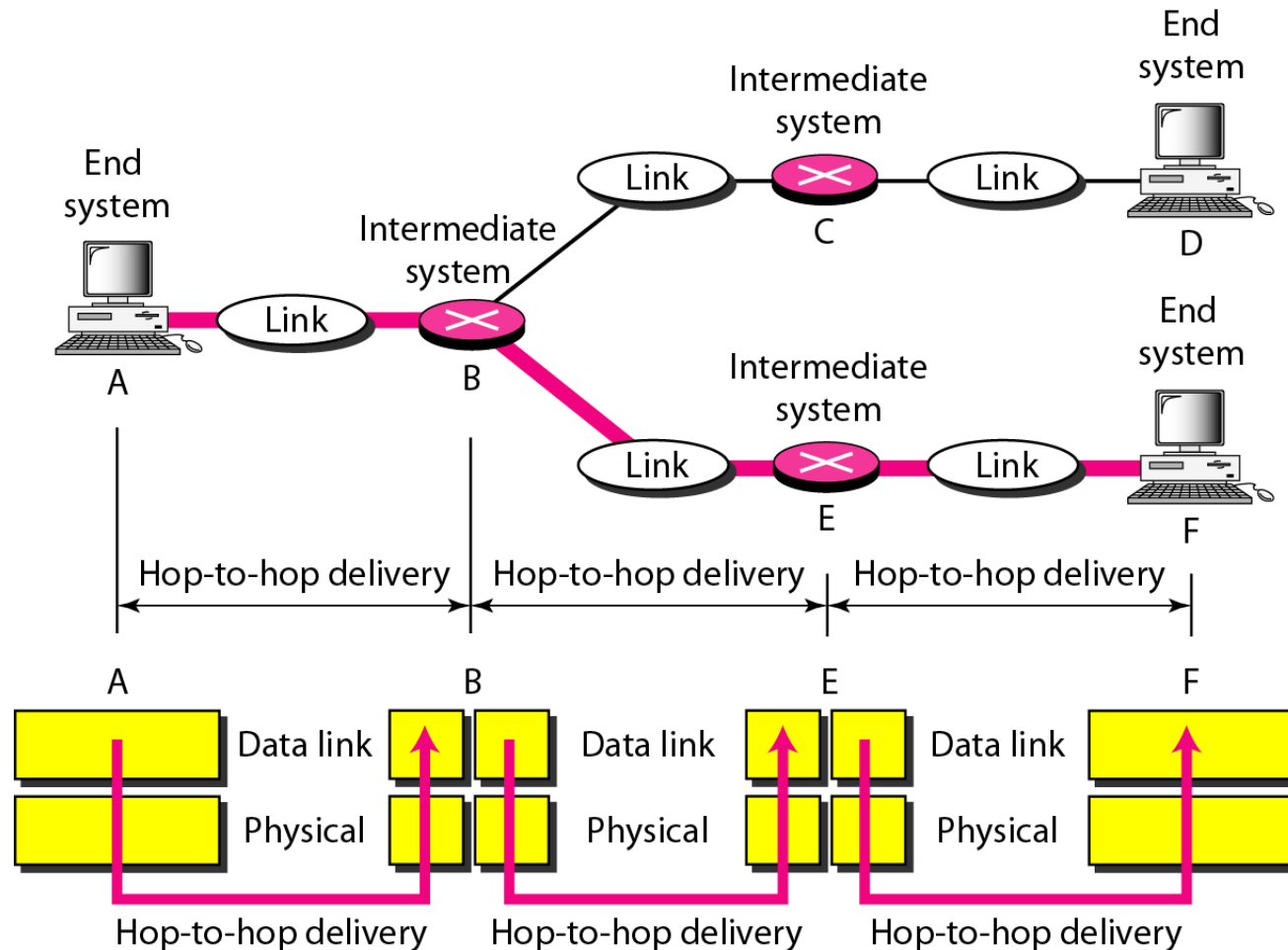
Flow Control: It is the technique through which the constant data rate is maintained on both the sides so that no data get corrupted. It ensures that the transmitting station such as a server with higher processing speed does not exceed the receiving station, with lower processing speed.

Error Control: is achieved by adding a calculated value CRC (Cyclic Redundancy Check) that is placed to the Data link layer's trailer which is added to the message frame before it is sent to the physical layer.

Access Control: When two or more devices are connected to the same communication channel, then the data link layer protocols are used to determine which device has control over the link at a given time.

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The OSI Architecture – Datalink layer

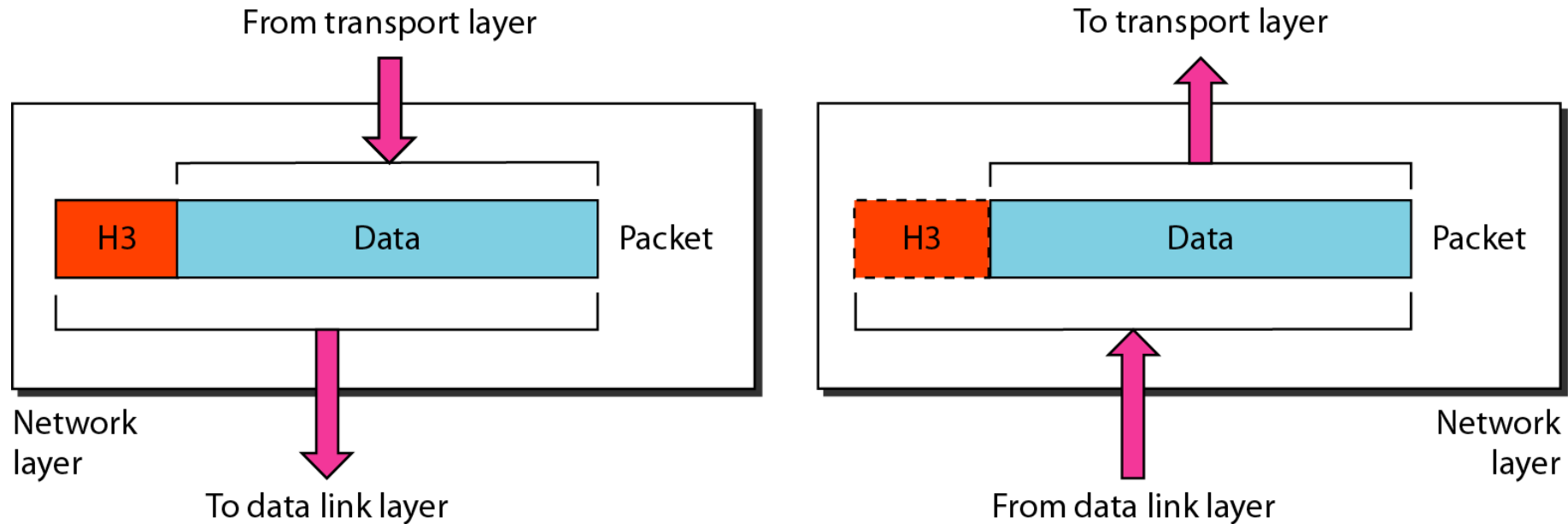


Network Layer: Functions

- **Internetworking:** It provides a logical connection between different devices.
- **Addressing:** A Network layer adds the source and destination address to the header of the frame. Addressing is used to identify the device on the internet(IPv4, IPv6)
- **Routing:** Routing is the major component of the network layer, and it determines the best optimal path out of the multiple paths from source to the destination.
- **Packetizing:** A Network Layer receives the packets from the upper layer and converts them into packets. This process is known as Packetizing. It is achieved by internet protocol (IP).

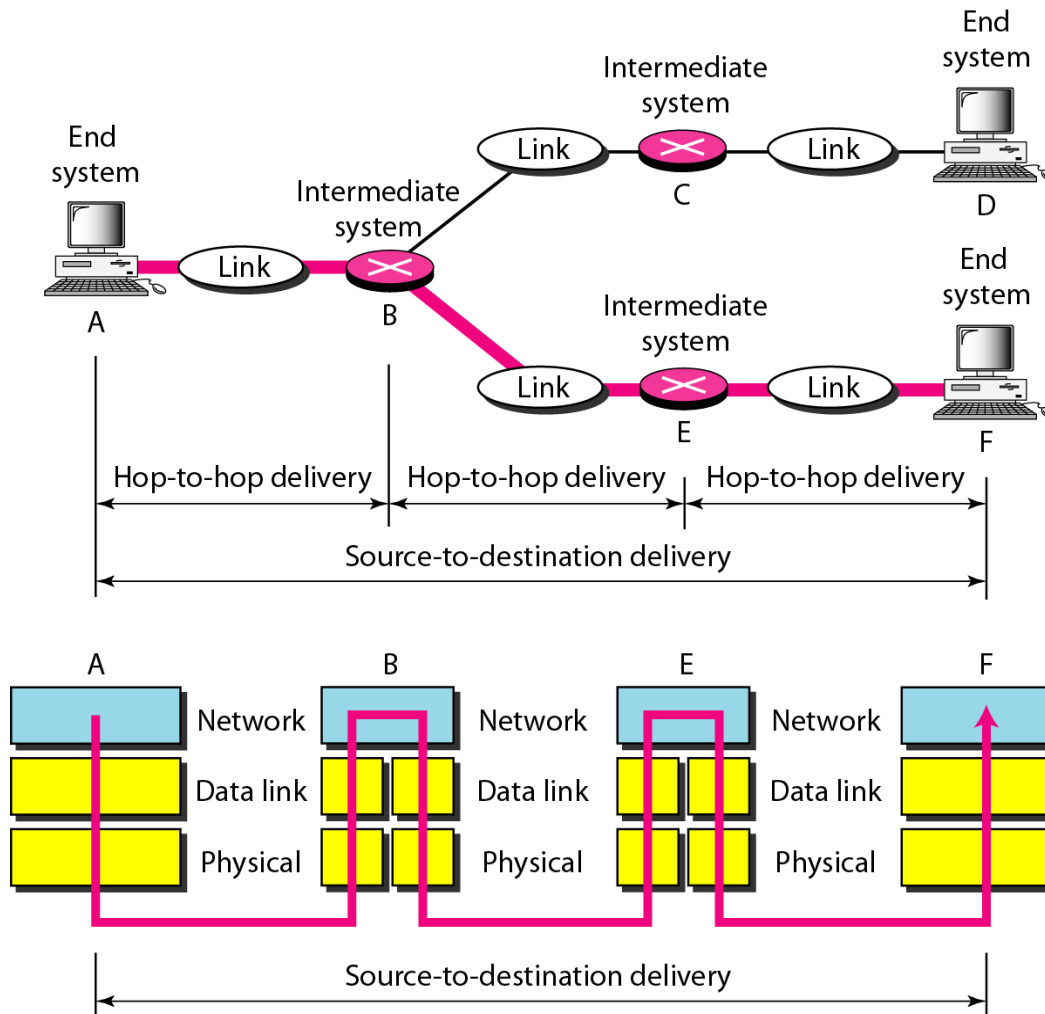
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The OSI Architecture – Network layer



Data Communications

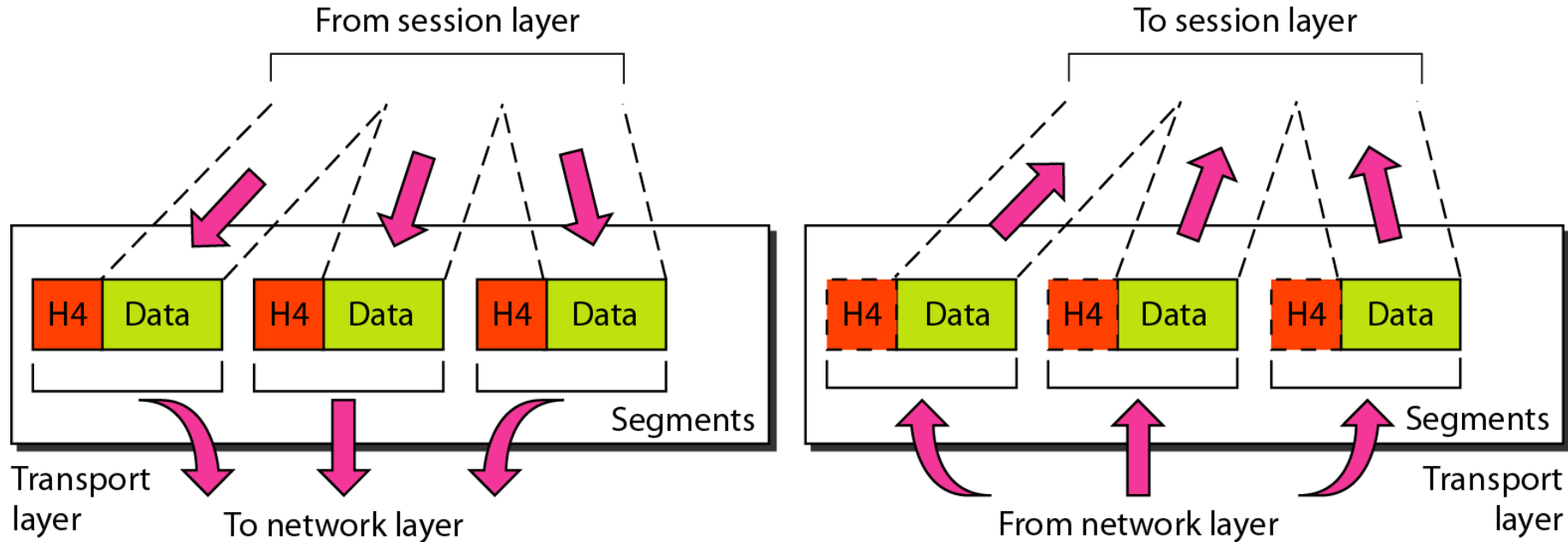
The OSI Architecture – Network layer



- The Transport layer ensures that messages are transmitted in the order in which they are sent and there is no duplication of data.
- The main responsibility of the transport layer is to transfer the data completely.
- It receives the data from the upper layer and converts them into smaller units known as segments.
- This layer can be termed as an end-to-end layer as it provides a point-to-point connection between source and destination to deliver the data reliably.
- **The two protocols used in this layer are: TCP and UDP**

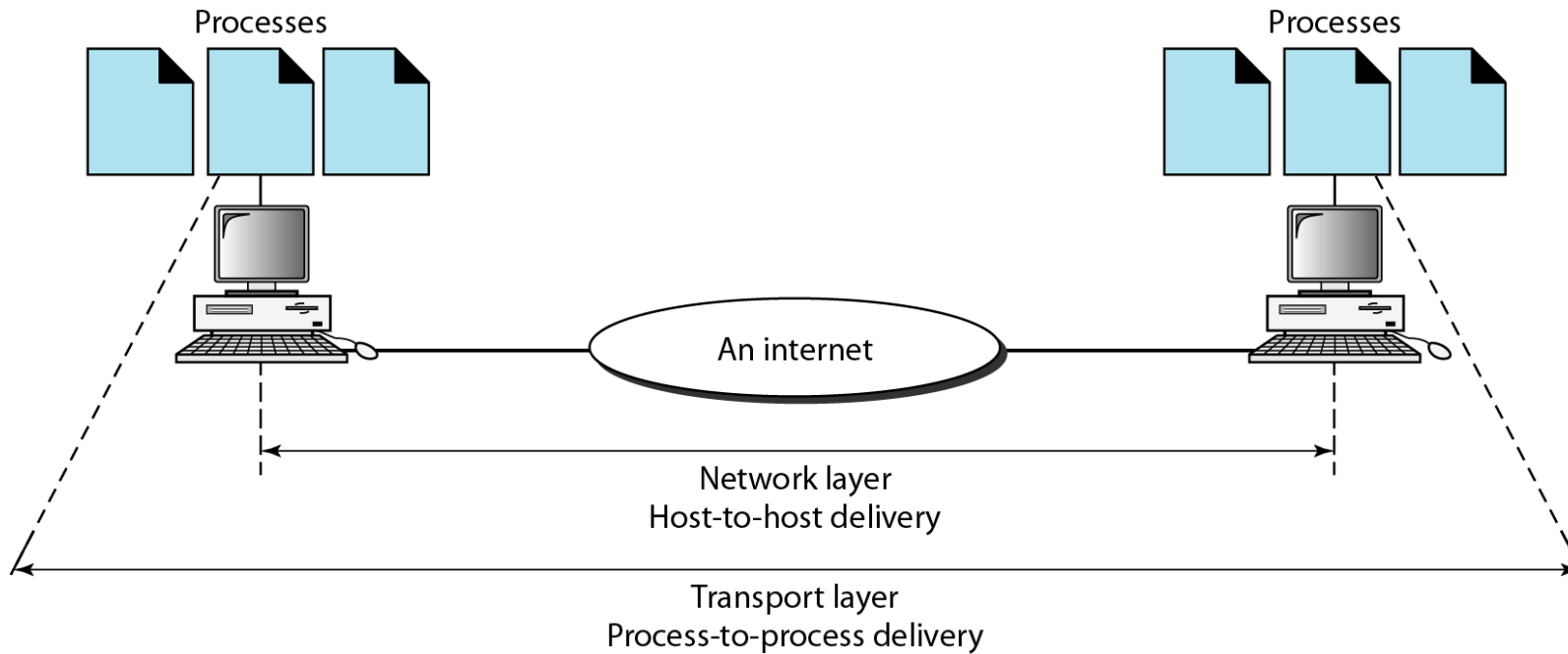
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The OSI Architecture – Transport layer



Data Communications

The OSI Architecture – Transport layer



- **Service-point addressing:** The responsibility of the network layer is to transmit the data from one computer to another computer
- **Segmentation and reassembly:** When the transport layer receives the message from the upper layer, it divides the message into multiple segments, and each segment is assigned with a sequence number that uniquely identifies each segment. At the destination, then the transport layer reassembles the message based on their sequence numbers.
- **Connection control:** Transport layer provides two services Connection-oriented service and connectionless service. In connectionless service each segment travels in different routes to reach the destination. In connection-oriented service, all the packets travel in the single route.
- **Flow control and Error control**

Session Layer:

The Session layer is used to establish, maintain and synchronizes the interaction between communicating devices.

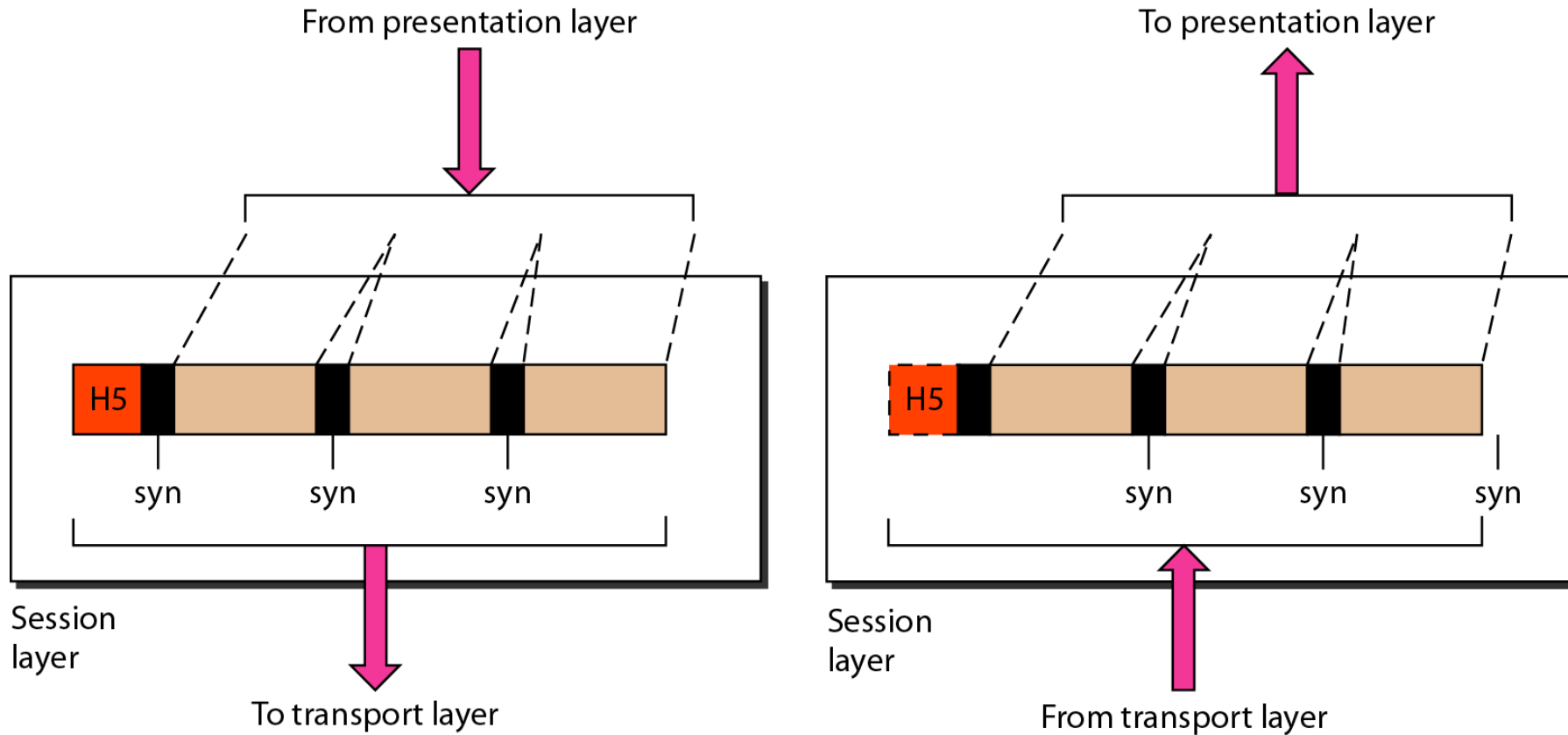
Functions:

Dialog control: Session layer acts as a dialog controller that it allows the communication between two processes which can be either half-duplex or full-duplex.

Synchronization: Session layer adds some checkpoints when transmitting the data in a sequence. If some error occurs in the middle of the transmission of data, then the transmission will take place again from the checkpoint. This process is known as Synchronization and recovery.

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The OSI Architecture – Session Layer

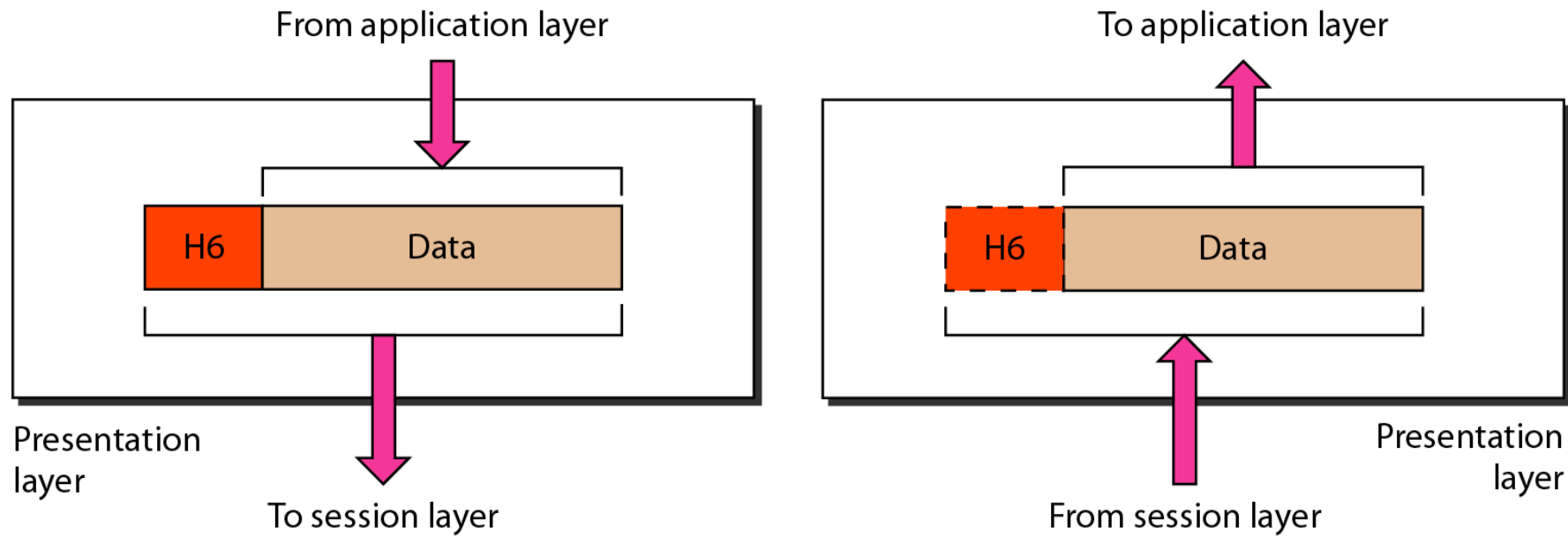


Presentation Layer:

- Presentation layer is mainly concerned with the syntax and semantics of the information exchanged between the two systems.
- This layer is a part of the operating system that converts the data from one presentation format to another.
- The Presentation layer is also known as the syntax layer.

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The OSI Architecture - Presentation Layer



Functions of Presentation layer:

- **Translation:** The processes in two systems exchange the information in the form of character strings, numbers and so on. Different computers use different encoding methods, the presentation layer handles the interoperability between the different encoding methods.
- **Encryption:** It is a process of converting the sender-transmitted information into another form and sends the resulting message over the network.
- **Compression:** Data compression is a process of compressing the data, i.e., it reduces the number of bits to be transmitted. Data compression is very important in multimedia such as text, audio, video.

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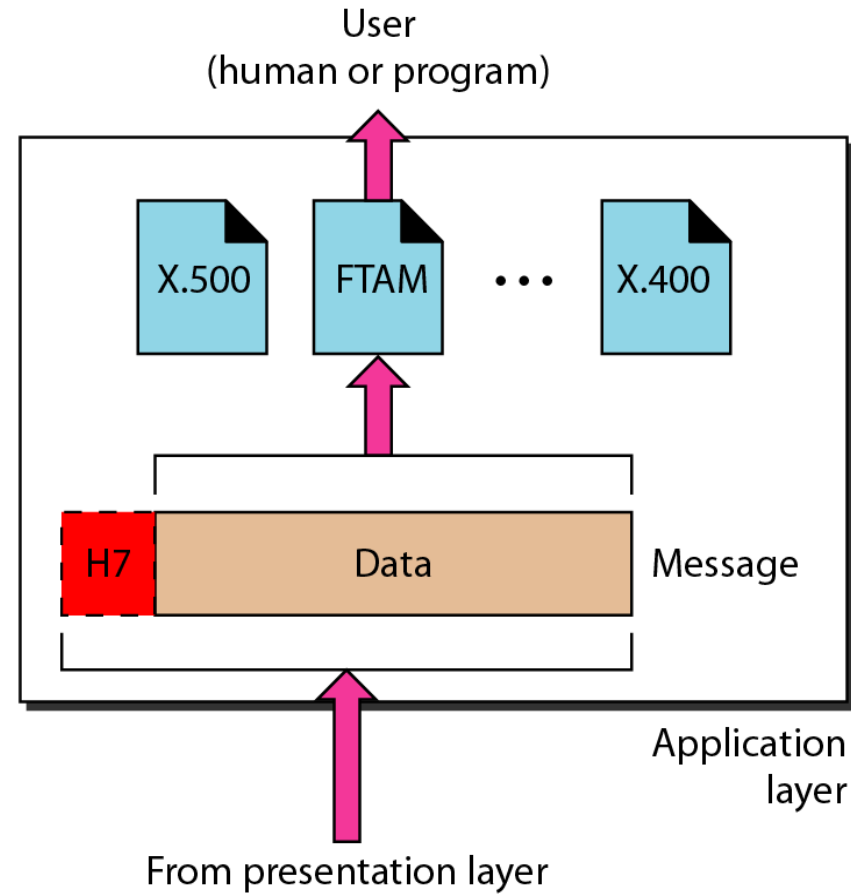
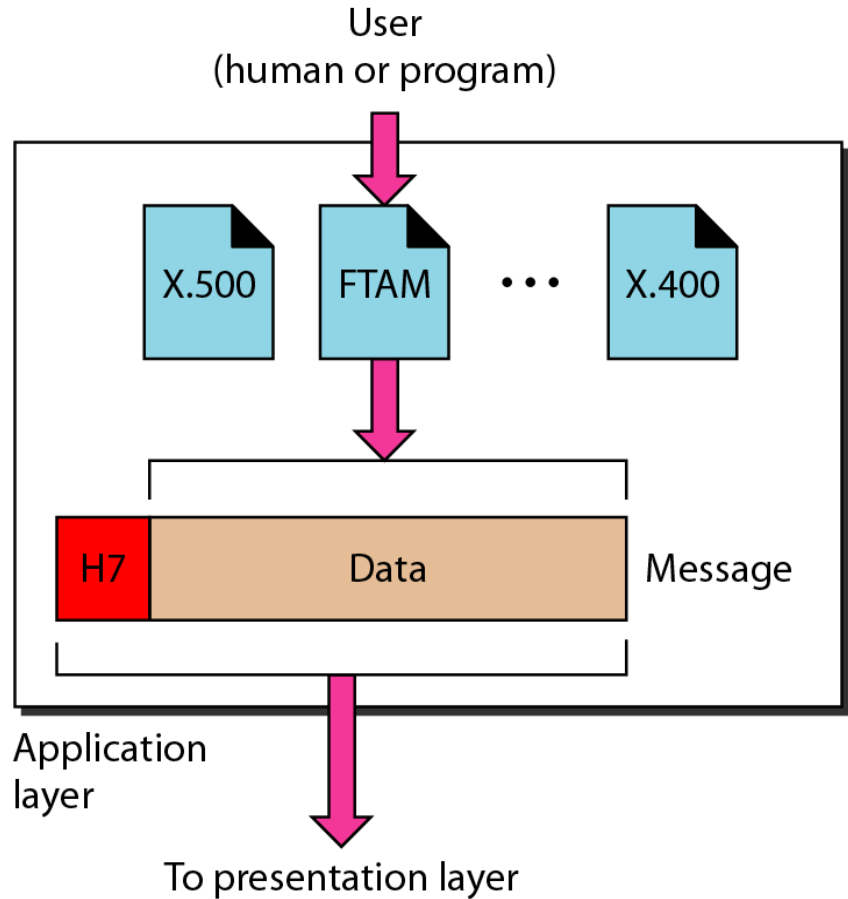
The OSI Architecture – Application Layer

Application Layer:

- **File transfer, access, and management (FTAM):** An application layer allows a user to access the files in a remote computer, to retrieve the files from a computer and to manage the files in a remote computer.
- **Mail services:** An application layer provides the facility for email forwarding and storage.
- **Directory services:** An application provides the distributed database sources and is used to provide that global information about various objects.

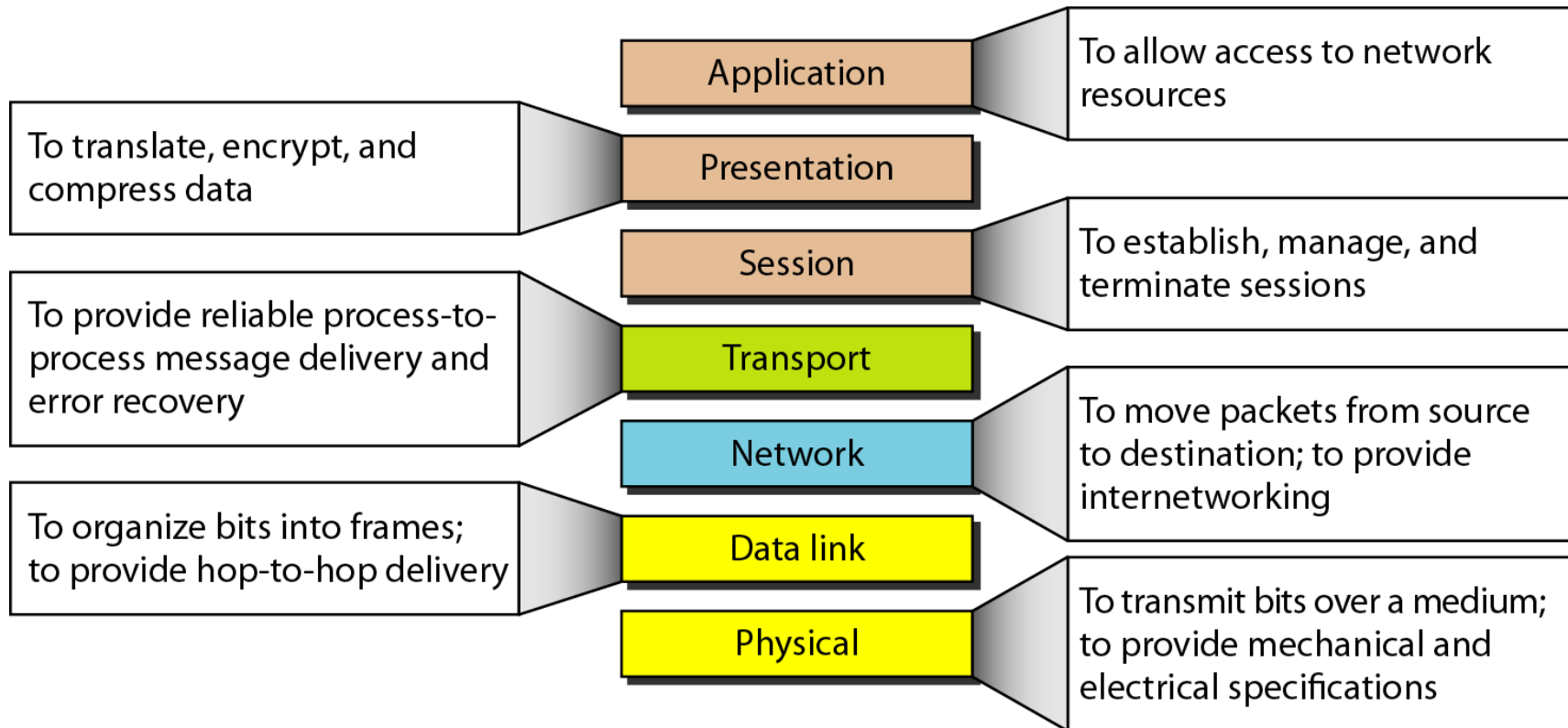
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The OSI Architecture – Application Layer



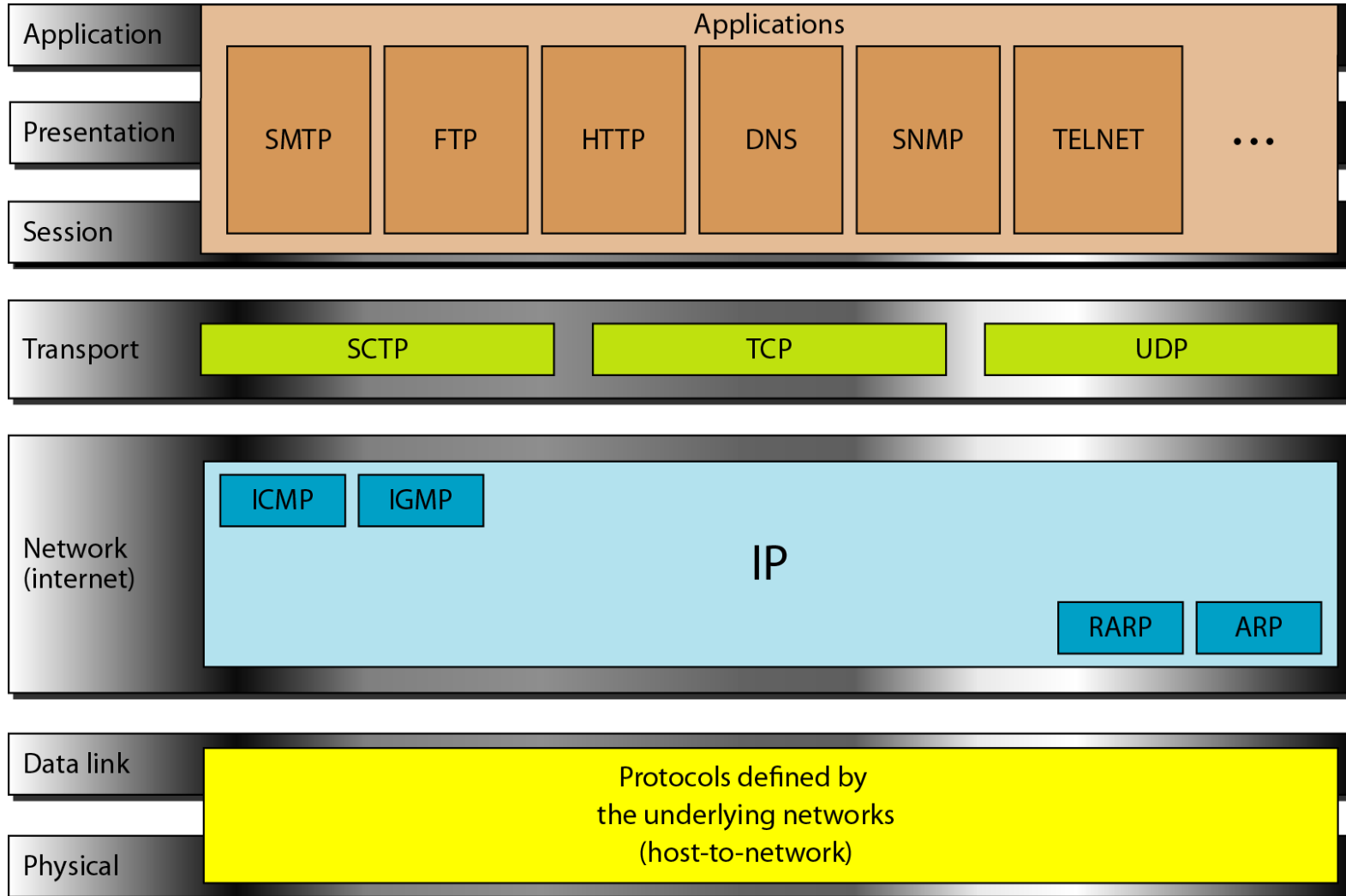
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The OSI Architecture –Summary



Data Communications

The OSI Architecture –Summary





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Data and Signals

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The physical layer's major function is to move data in the form of electromagnetic signals over a transmission medium.

- Data can be **analog or digital**.
- Analog data refers to information which is continuous.
- Example, an Analog clock which is having hour, minute, and second hands that gives information in a continuous form; the movement of the hands are continuous.
- Analog data take on continuous values.

Digital Data

- Digital data refers to information which has discrete states.
- Example, a digital clock which reports the hours and minutes that will change suddenly from 10:05 to 10:06.
- Digital data take on discrete values
- When someone speaks, an analog wave is created in the air. This is captured by microphone and converted into analog or digital signal.

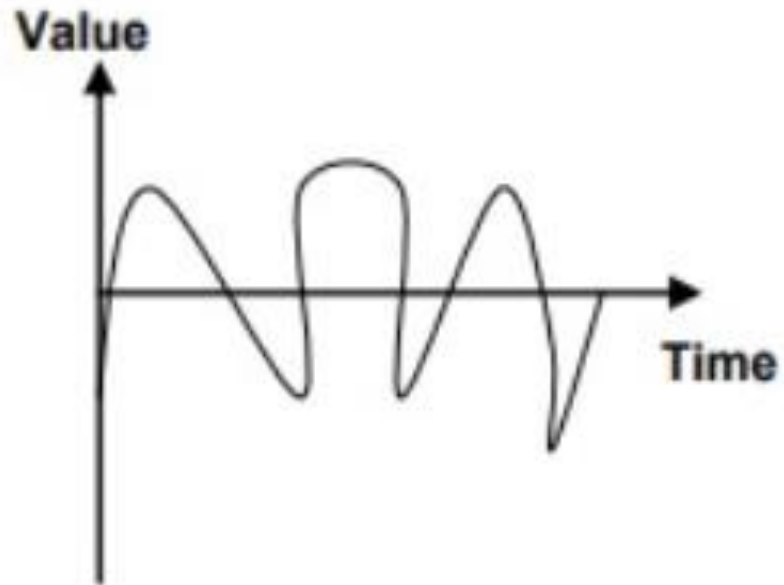
Analog signal

Analog signals have infinitely so many levels of intensity over a period of time.

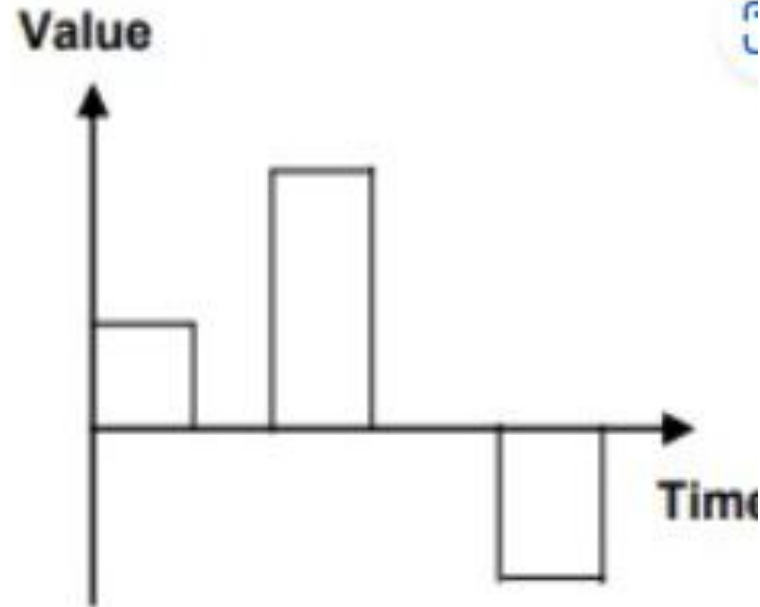
When the wave moves from value A to value B, it passes through and it includes an infinite number of values along its path.

Digital Signal

Digital signals can have only a limited number of defined values. Each value can be any number, it is often 0 or 1.



a. Analog Signal



b. Digital Signal

Periodic signal

- Periodic signals are a type of electrical signals which repeat their behavior after a certain interval of time known as time period.
- It is also known as a cyclic function.
- The completion of one full pattern is called a cycle.

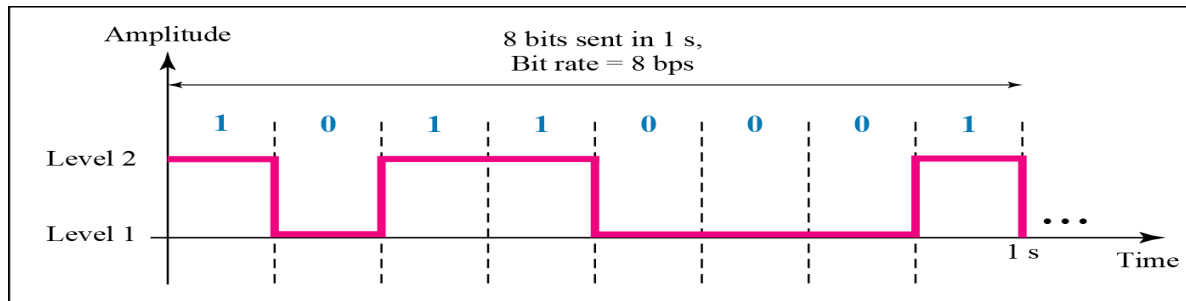
Non Periodic or aperiodic signal

- A non periodic signal changes without exhibiting a pattern or cycle.
- Analog and digital signals can be either periodic or non periodic

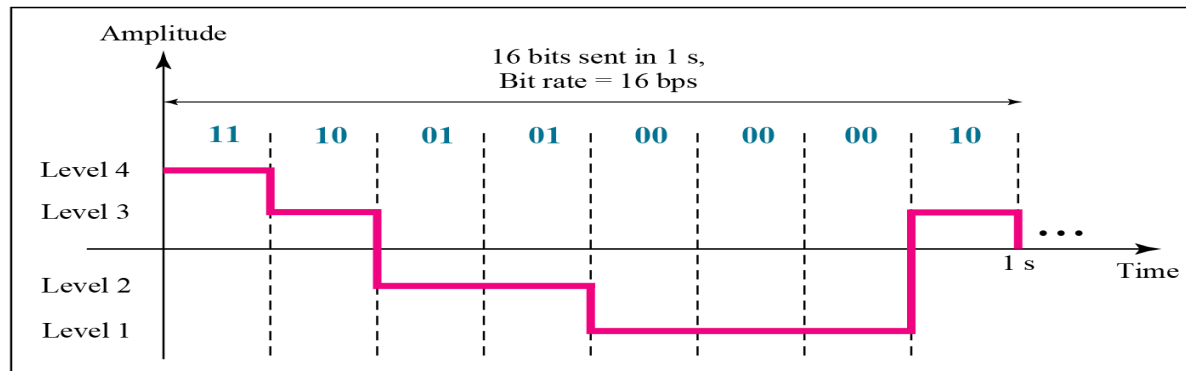
Data Communications

Digital Signals

- Information can be represented by a digital signal.
- A digital signal can have more than two levels.



a. A digital signal with two levels



b. A digital signal with four levels

To calculate the number of bits needed per level,

$$\text{Number of bits per level} = \log_2 8 = 3$$

Example 1: A digital signal has eight levels. How many bits are needed per level?

Each signal level is represented by 3 bits.

Example 2: A digital signal has nine levels. How many bits are needed per level? Each signal level is represented by 3.17 bits. However, this answer is not realistic. The number of bits sent per level needs to be an integer as well as a power of 2. For this example, 4 bits can represent one level.

Bit rate:

- Bit rate is the rate at which data is processed or transferred.
- It is the number of bits sent in 1 second(bps)
- The unit of measuring bit rate is seconds. It ranges from bps for small units and Kbps, Mbps, and Gbps for higher bit rates.
- Bit rate is also known as data rate.

Bit rate – Problems

Q1: Assume we need to download text documents at the rate of 100 pages per second. What is the required bit rate of the channel?

Solution:

Assume the page is an average of 24 lines with 80 characters in each line.

One character requires 8 bits.

The bit rate = $100 * 24 * 80 * 8 = 1,536,000 \text{ bps} = 1.536 \text{ Mbps}$

Bit rate – Problems

- **Q2:** A digitized voice channel, as we will see in Chapter 4, is made by digitizing a 4-kHz bandwidth analog voice signal. We need to sample the signal at twice the highest frequency (two samples per hertz). We assume that each sample requires 8 bits. What is the required bit rate?

Solution

- The bit rate can be calculated as

$$\text{bit rate} = 2 * 4000 * 8 = 64000 \text{ bps} = 64 \text{ kbps}$$

Bit rate – Problems

- **Q3:** What is the bit rate for high-definition TV (HDTV)?

Solution

- HDTV uses digital signals to broadcast high quality video signals. The HDTV screen is normally a ratio of 16 : 9. There are 1920 by 1080 pixels per screen, and the screen is renewed 30 times per second. Twenty-four bits represents one color pixel.

The bit rate can be calculated as

$$\text{bit rate} = 1920 * 1080 * 30 * 24 = 1,492,992,000 \text{ bps} = 1.5 \text{ Gbps}$$

The TV stations reduce this rate to 20 to 40 Mbps through compression.

Bit length:

A bit length is the distance a bit occupies on the transmission medium

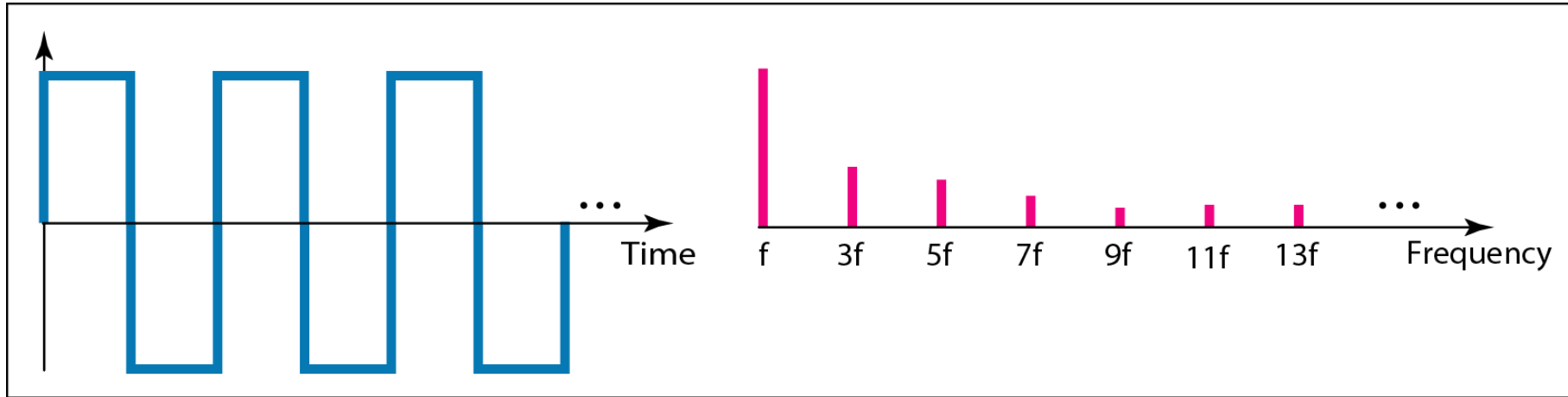
$$\text{Bit length} = \text{propagation speed} * \text{bit duration}$$

Bandwidth:

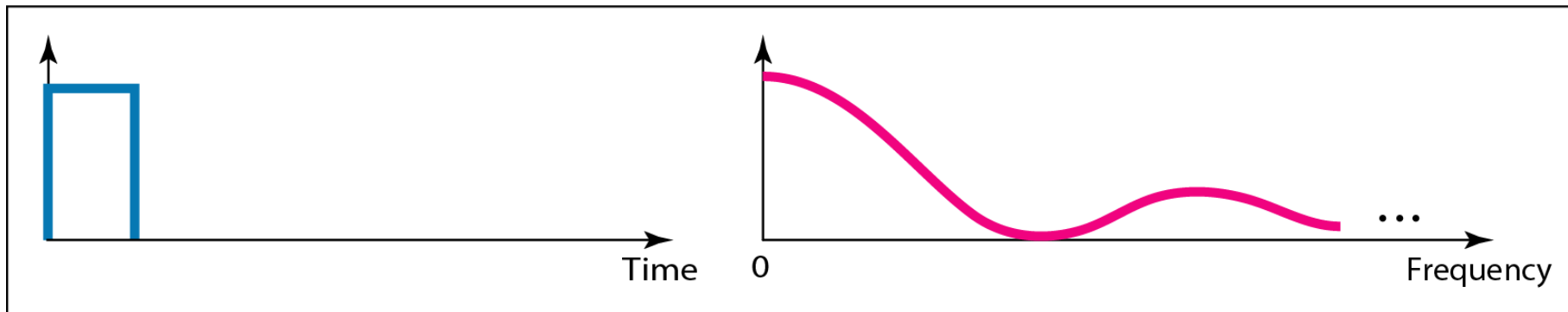
- The quantity of data that may be sent through a specific network connection
- It is expressed in bits per second (bps) or megabits per second (Mbps).
- A network connection's speed is determined by its Bandwidth.
- The faster a user can access and download data, the higher the Bandwidth.

If an Internet connection had a bandwidth of 56 Kbps (kilobits per second). How many bits per second could that transfer?

- A. 560 bps
- B. 5600 bps
- C. 56000 bps
- D. 560000 bps



a. Time and frequency domains of periodic digital signal



b. Time and frequency domains of nonperiodic digital signal



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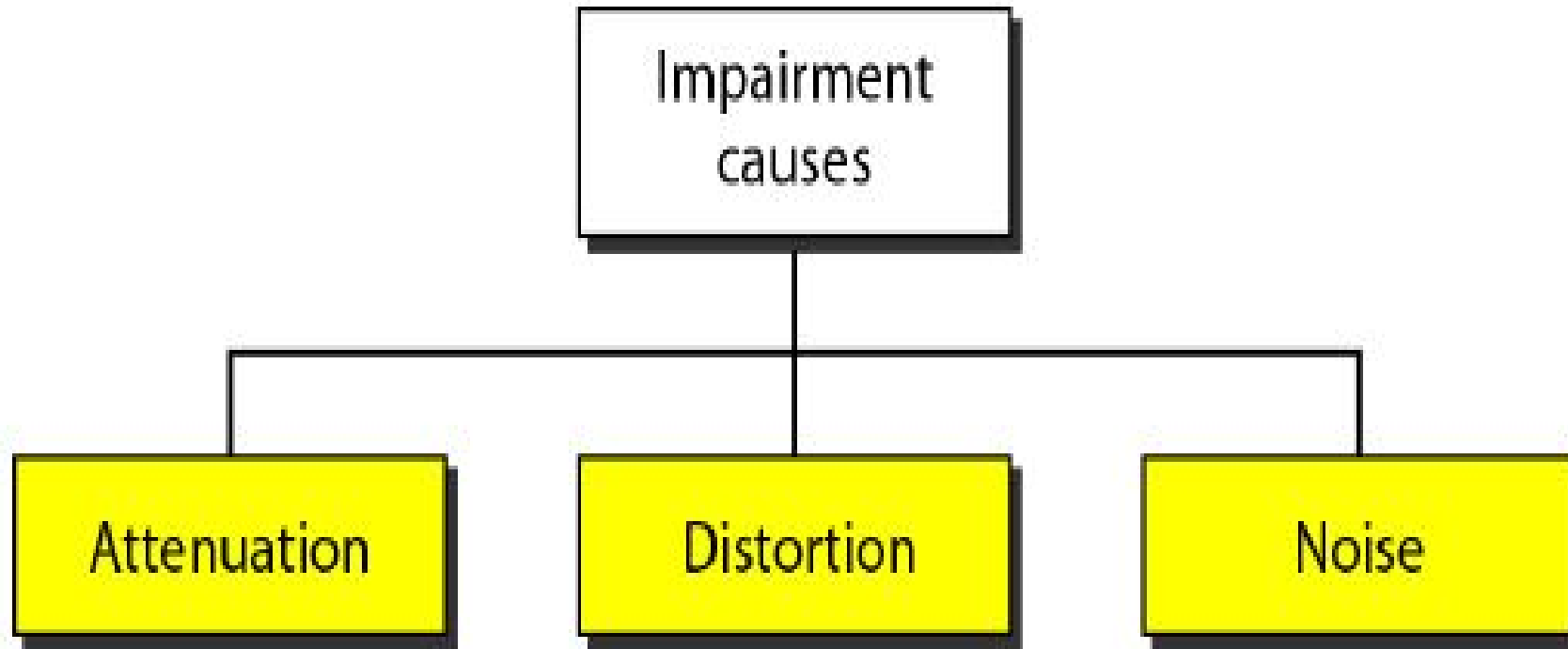


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- Transmission impairment occurs when the received signal is different from the transmitted signal.
- In Analog signals, due to transmission impairment the resulting received signal gets different amplitude or the shape.
- In the digitally transmitted signals at the receiver side the changes occurs in bits (0's or 1's).



Attenuation

- Attenuation means loss of energy.
- When a signal travels through a medium it loses its energy in overcoming the resistance of the medium.
- The decrease in signal strength, by which the received signal will be difficult to receive at the receiver end.
- To compensate the loss, amplifiers (Analog signals) or repeaters (digital signals) are used.
- To show the loss or gain of energy the unit “decibel” is used.

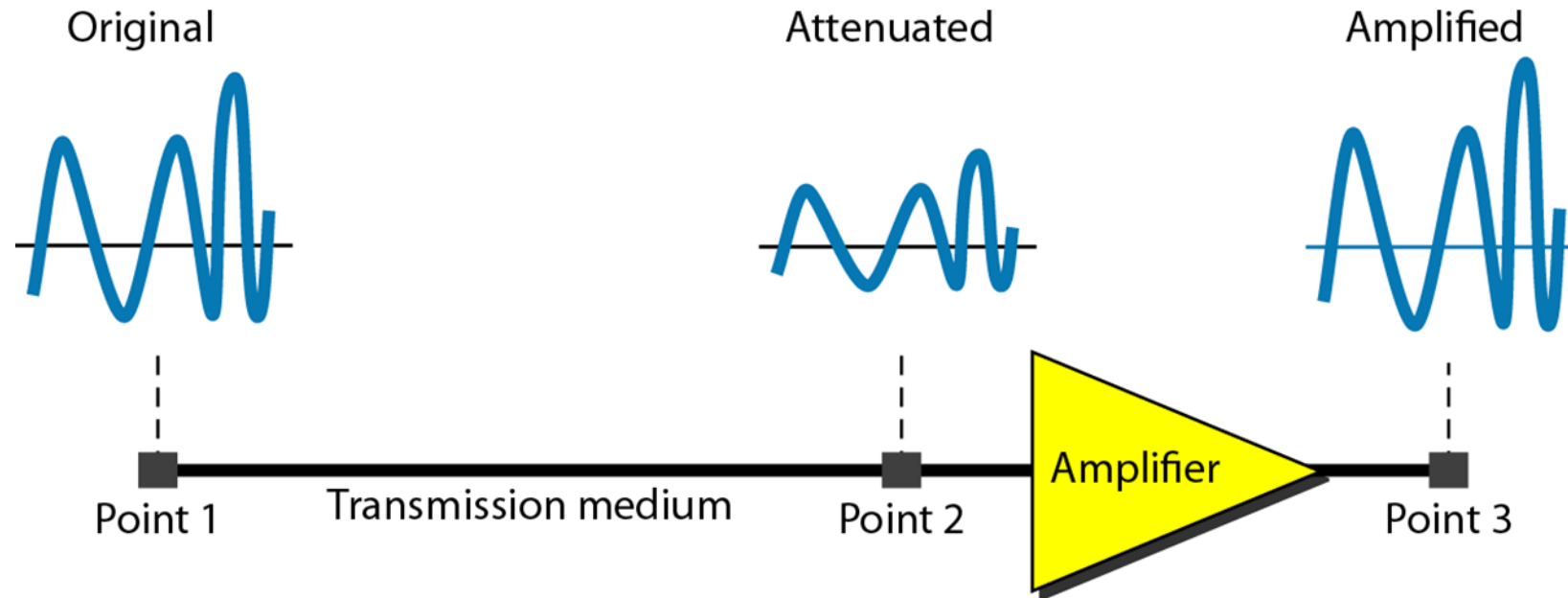
$$\text{dB} = 10\log_{10}P_2/P_1$$

P_1 - input signal, P_2 - output signal

Data Communications

Transmission Impairments

Attenuation



Attenuation

Numerical Problem 1:

Suppose a signal travels through a transmission medium and its power is reduced to one-half. This means that P_2 is $(1/2)P_1$. In this case, the attenuation (loss of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5 P_1}{P_1} = 10 \log_{10} 0.5 = 10(-0.3) = -3 \text{ dB}$$

A loss of 3 dB (−3 dB) is equivalent to losing one-half the power.

Attenuation

Numerical Problem 2:

A signal travels through an amplifier, and its power is increased 10 times. This means that $P_2 = 10P_1$. In this case, the amplification (gain of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{10P_1}{P_1} = 10 \log_{10} 10 = 10(1) = 10 \text{ dB}$$

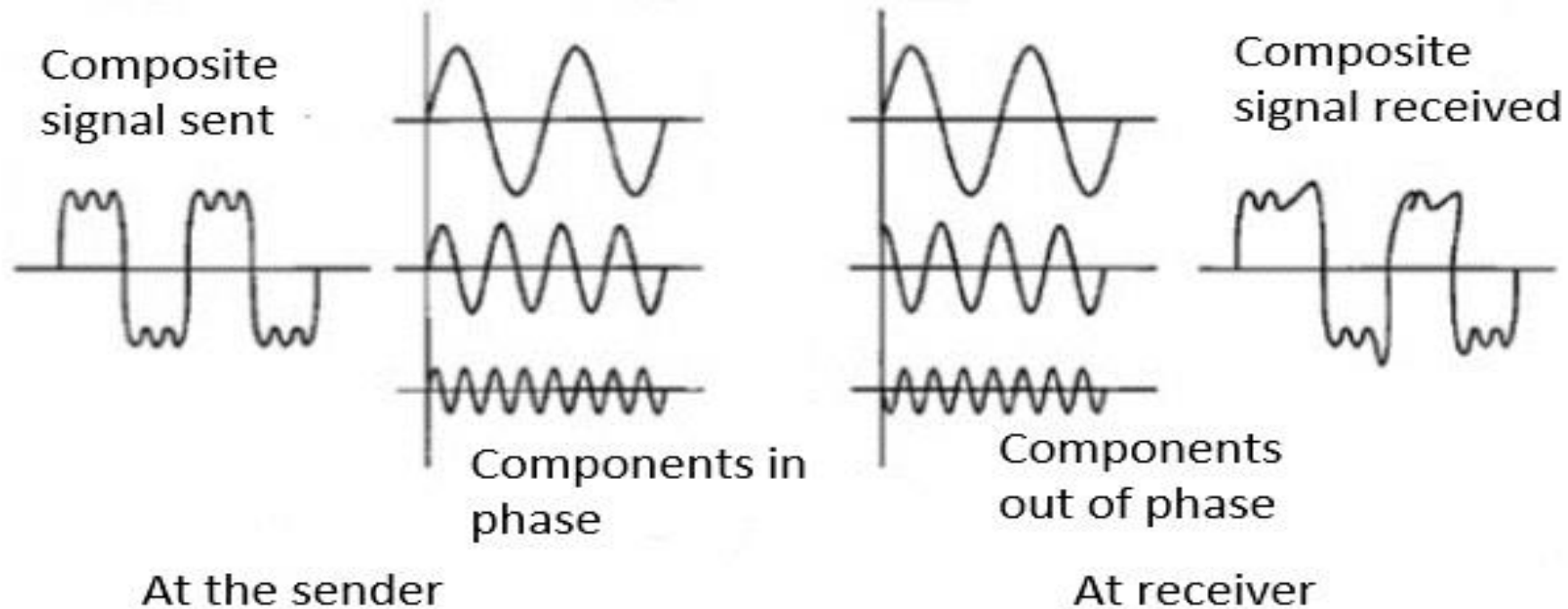
Distortion

- Distortion in a composite signal occurs when its shape or form changes during transmission due to differing propagation speeds of its constituent frequencies.
- Analog signals, multi-frequency transmissions.
 - **Amplitude Distortion:** Uneven amplification.
 - **Delay Distortion:** Different frequency components arrive at different times.
- A composite signal has various frequency components in it and each frequency component has some time constraint which makes a complete signal.
- While transmitting a composite signal, a certain delay happens between the frequencies components, then there may be the chance that the frequency component will reach the receiver end with a different delay constraint from its original which leads to the change in shape of the signal.
- **Solution:** Equalizers are used to balance signal delays.

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Transmission Impairments

Distortion



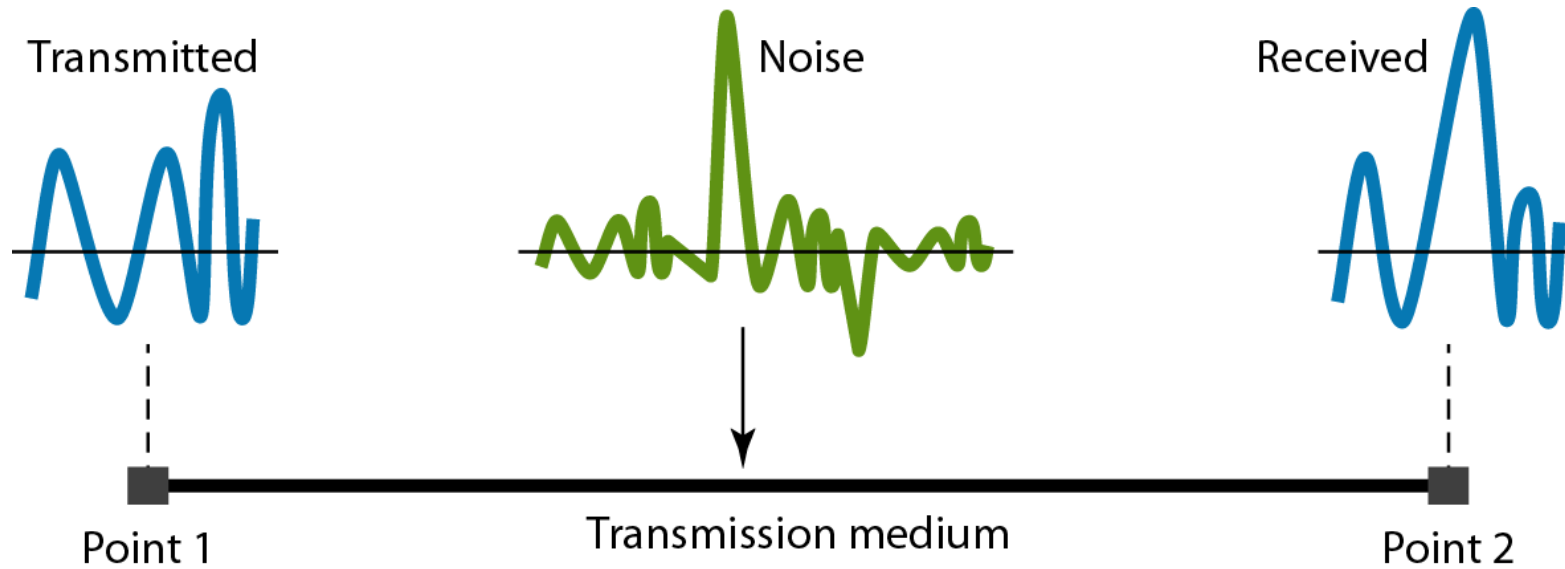
Noise

- Any unwanted signal gets added to the transmitted signal
- **Thermal noise** is the random motion of electrons in a wire, that creates an extra signal not sent by transmitter
- **Induced noise** comes from sources like motor and other appliances.
- **Crosstalk** is the effect of one wire on the other.
- **Impulse noise** is a spike – signal with high energy in a very short time that comes from power lines, lightning.

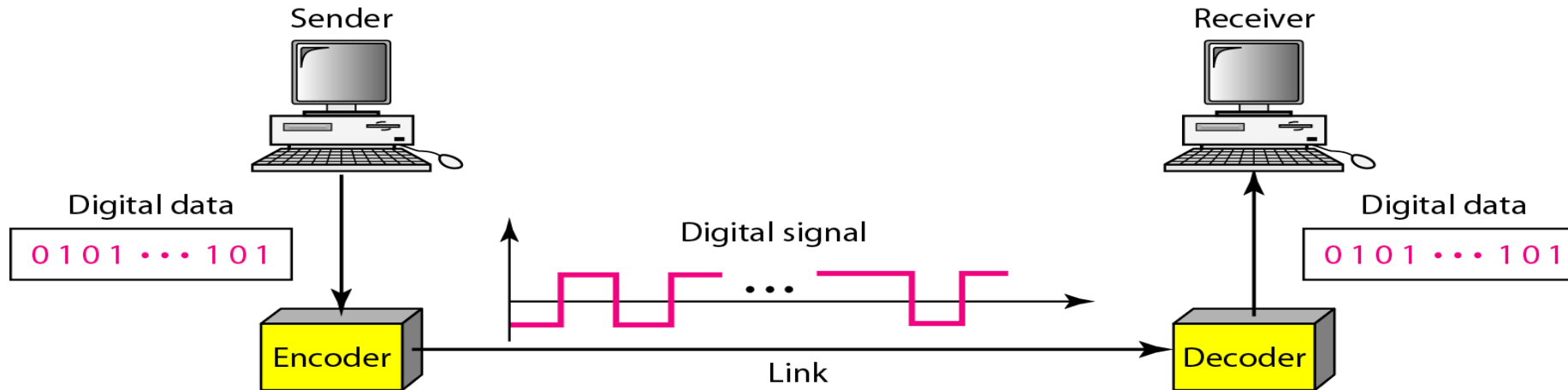
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Transmission Impairments

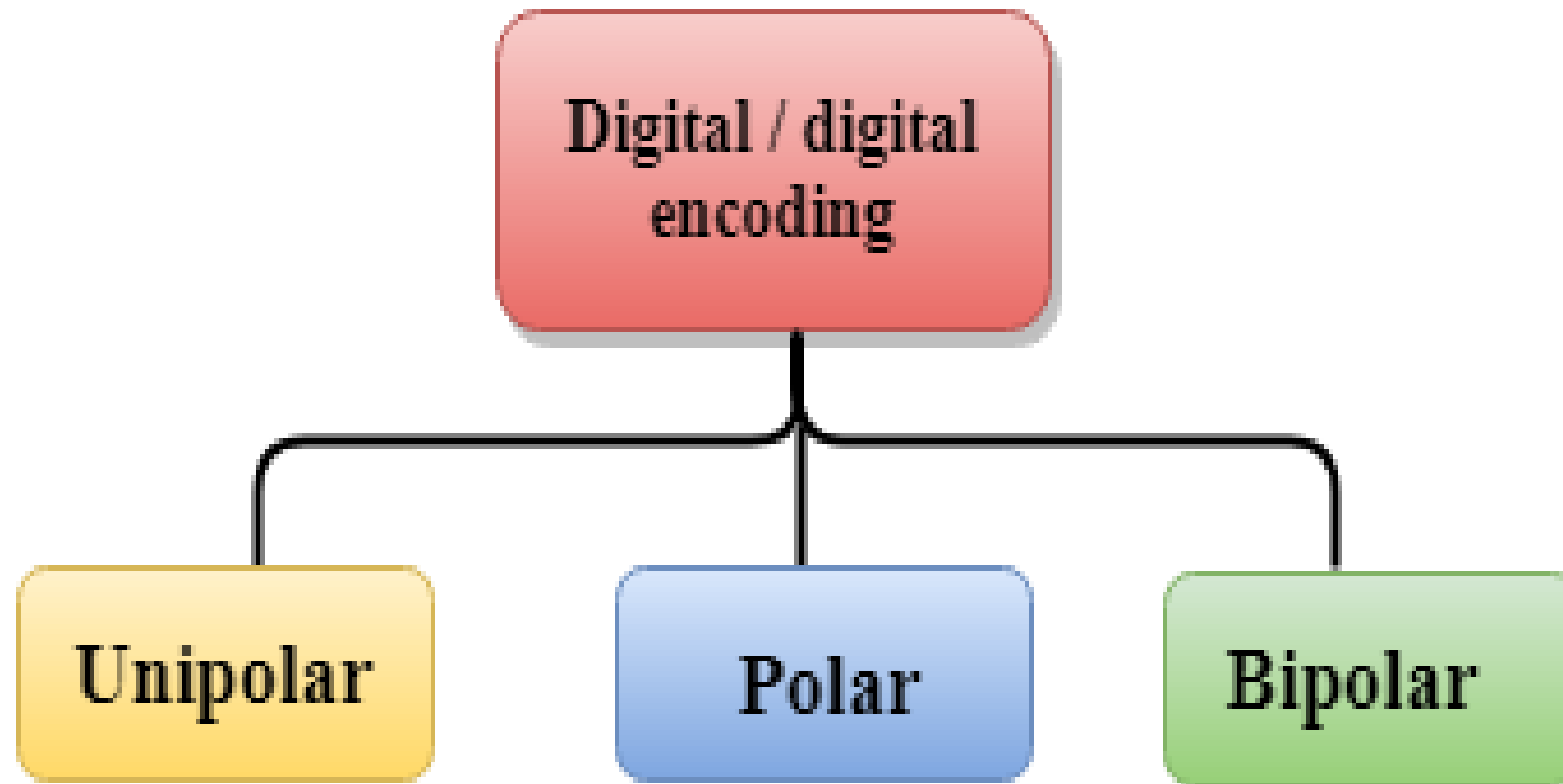
Noise



- Digital-to-digital encoding means conversion of digital data into digital signal and vice versa.
- This conversion can be done by **line coding** or **block coding** methods.
- The process of converting digital data to digital signal is line coding.
- Digital data – found in binary form, Digital signal – denoted by discrete signal



Digital-to-digital encoding is divided into three categories:

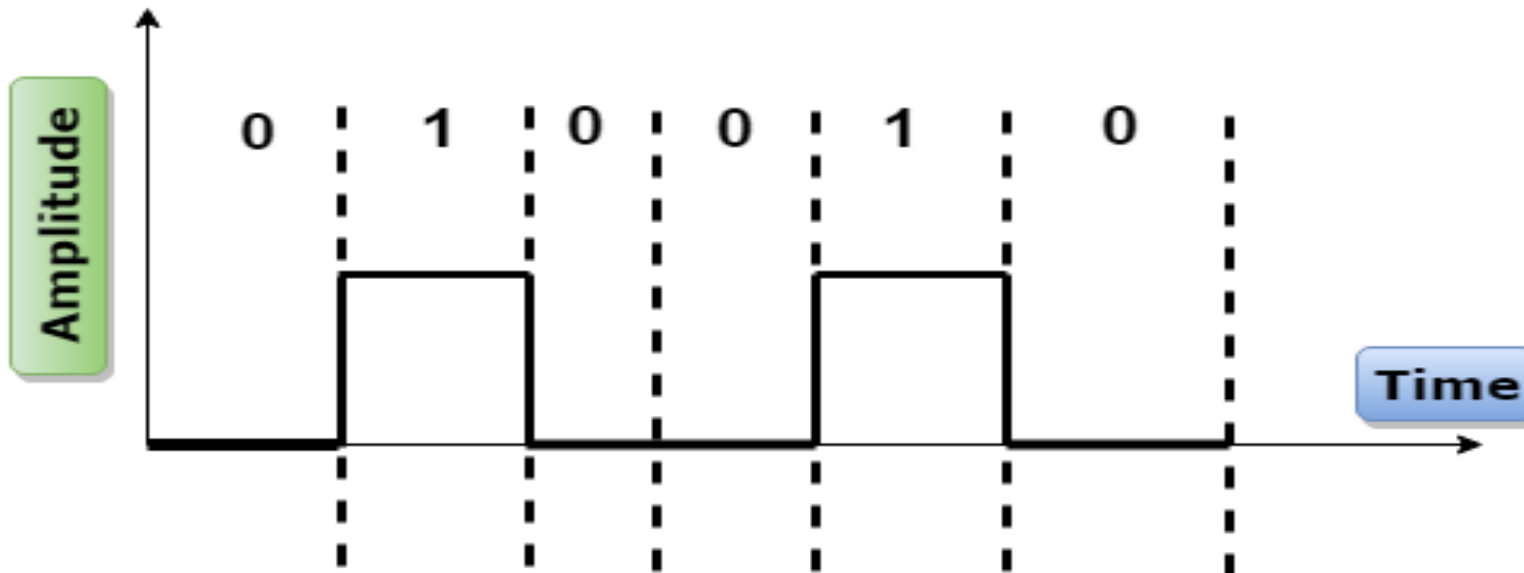


Unipolar

- Digital transmission system sends the voltage pulses over the link such as wire or cable.
- one voltage level represents 0, and another voltage level represents 1.
- The polarity of each pulse determines whether it is positive or negative.
- This method uses only one polarity.
- In Unipolar encoding, the polarity is assigned to the 1 binary state.
- 1s are represented as a positive value and 0s are represented as a zero value.

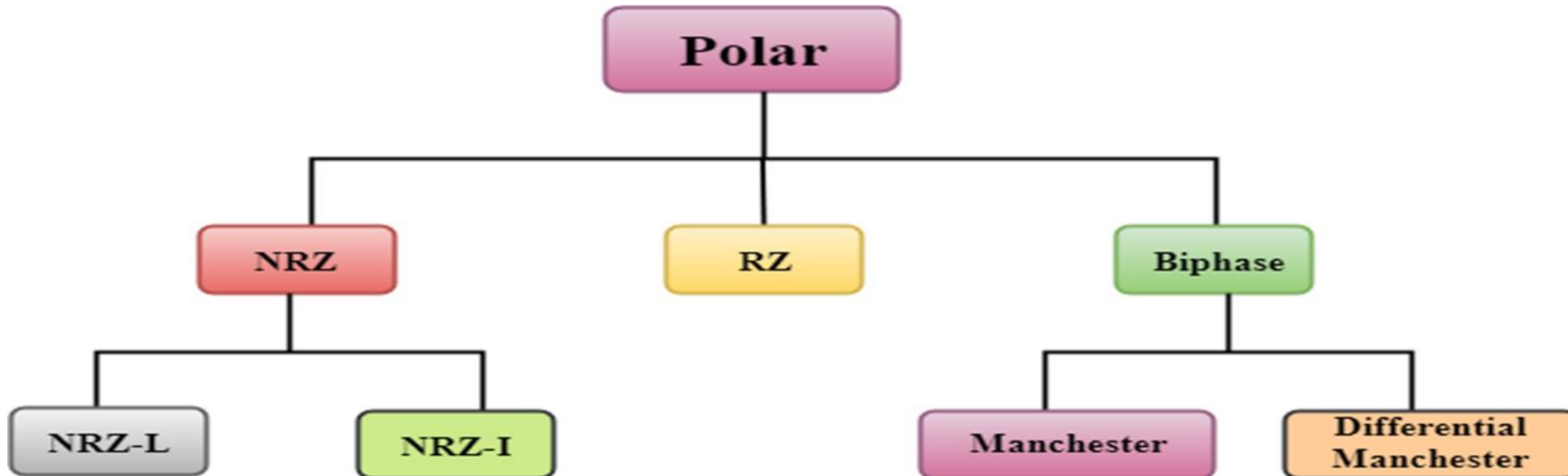
Unipolar

- Here, '1' is considered as a high voltage and '0' is considered as a zero voltage.
- Unipolar encoding is simpler and inexpensive to implement.



Polar

- Polar encoding uses two voltage levels: positive and negative.
- There are different categories of Polar encoding schemes.



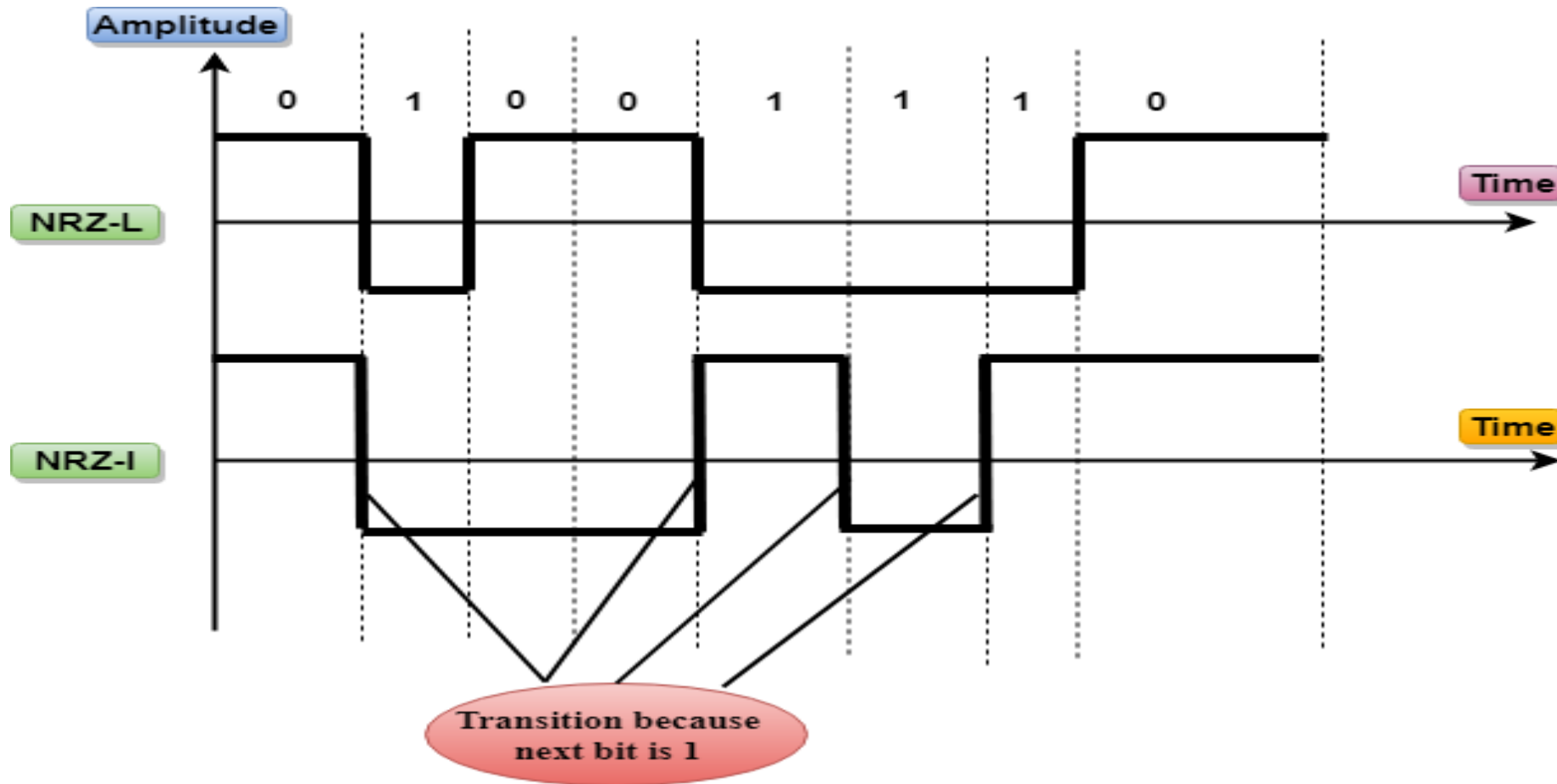
Polar – NRZ (Non-return to zero)

- In NRZ encoding, the level of the signal can be represented either positive or negative.
- **NRZ-L:** the level of the signal depends on the state of the bit. If a bit is 0 or 1, then their voltages will be positive and negative respectively.
- **NRZ-I:** NRZ-I is an inversion of the voltage level that represents 1 bit. In the NRZ-I encoding scheme, a transition occurs between the positive and negative voltage that represents 1 bit. In this scheme, 0 bit represents no change and 1 bit represents a change in voltage level.

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Digital-to-digital Conversion

- NRZ



RZ(Return to zero)

RZ is an encoding scheme that provides three values,

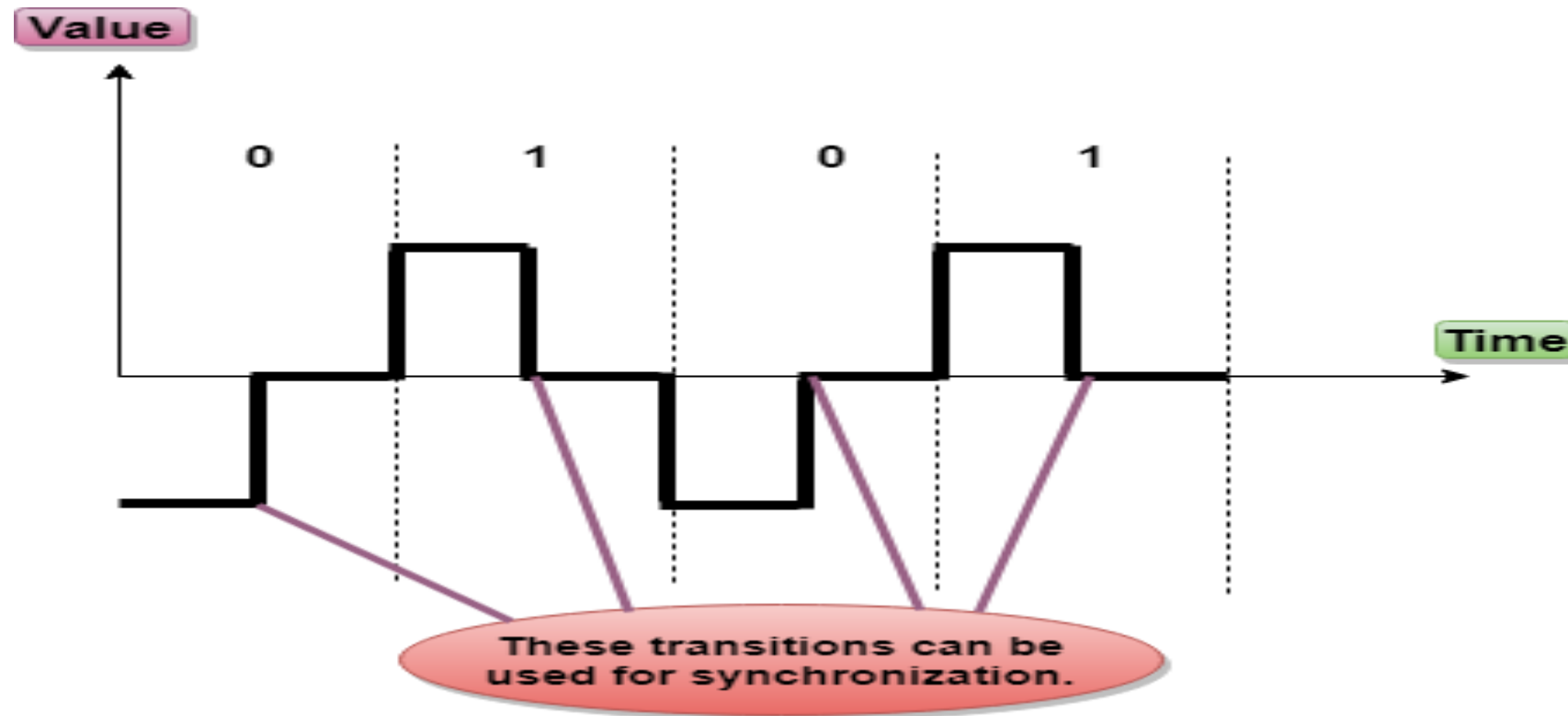
- positive voltage represents 1,
 - the negative voltage represents 0,
 - zero voltage represents none.
-
- In RZ scheme, halfway through each interval, the signal returns to zero.
-
- In RZ scheme, 1 bit is represented by **positive-to-zero** and 0 bit is represented by **negative-to-zero**.

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Digital-to-digital Conversion

Disadvantage of RZ:

- It performs two signal changes to encode one bit that acquires more bandwidth.





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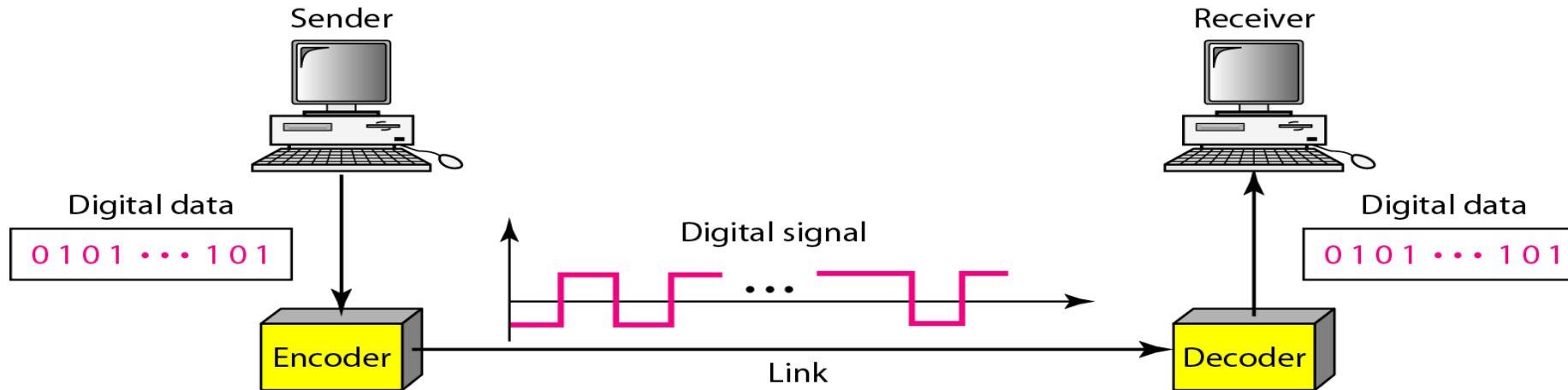
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Digital-to-Digital Conversion

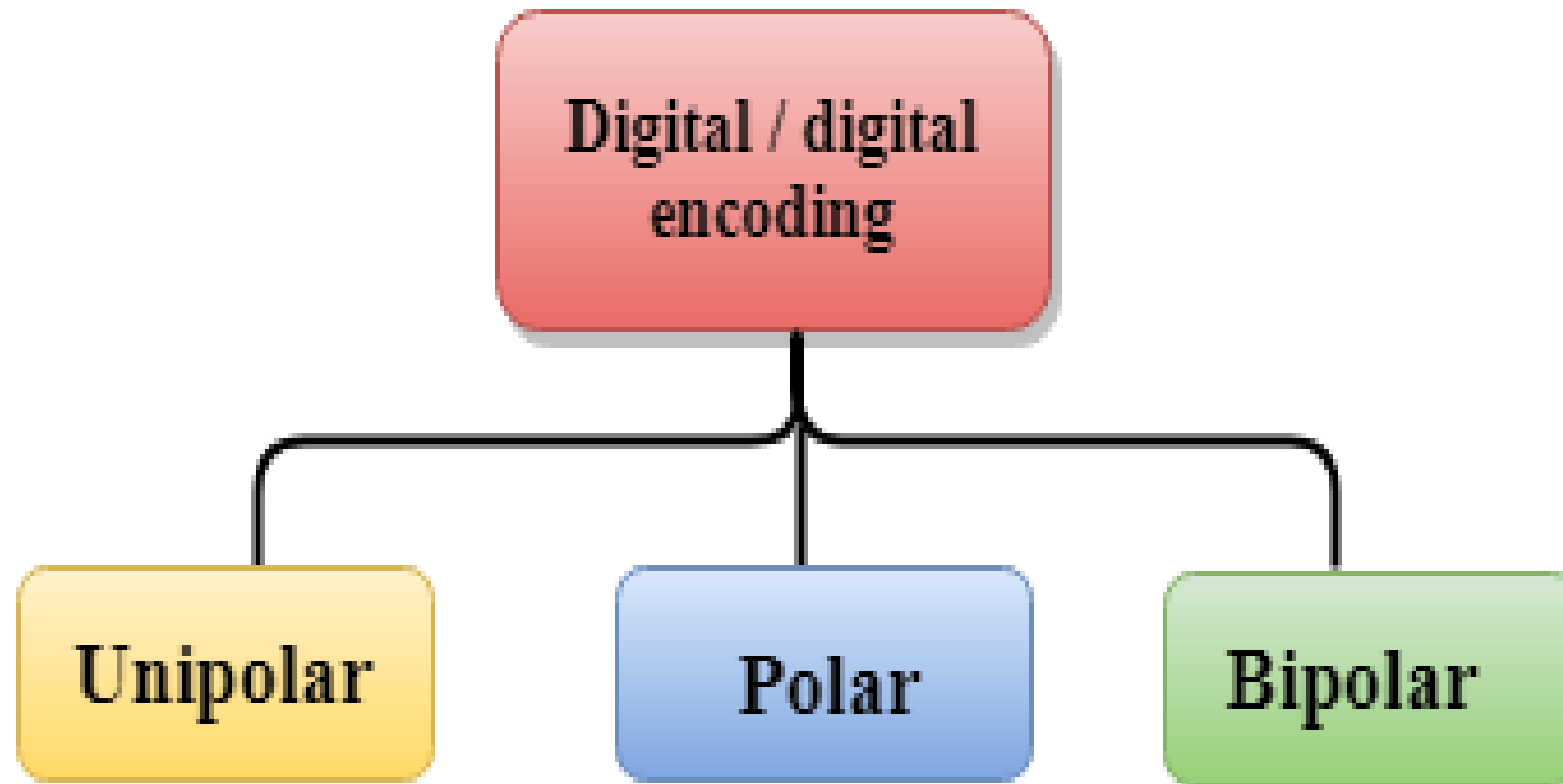
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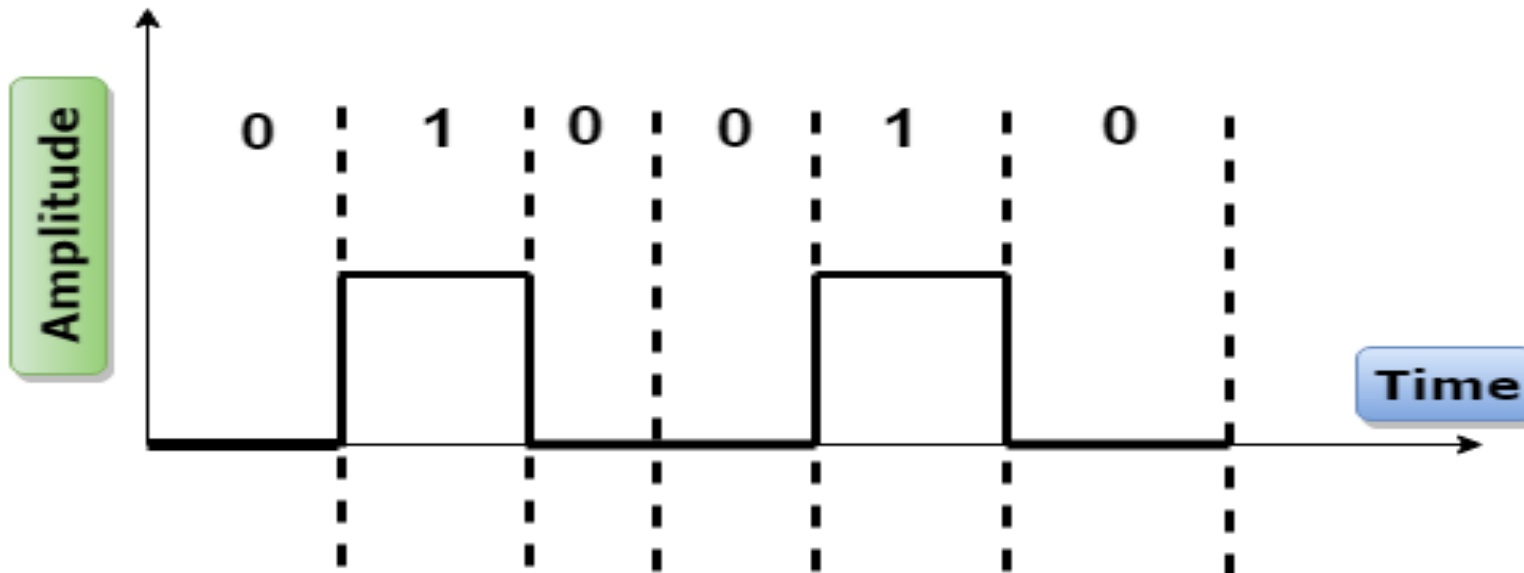


Unipolar

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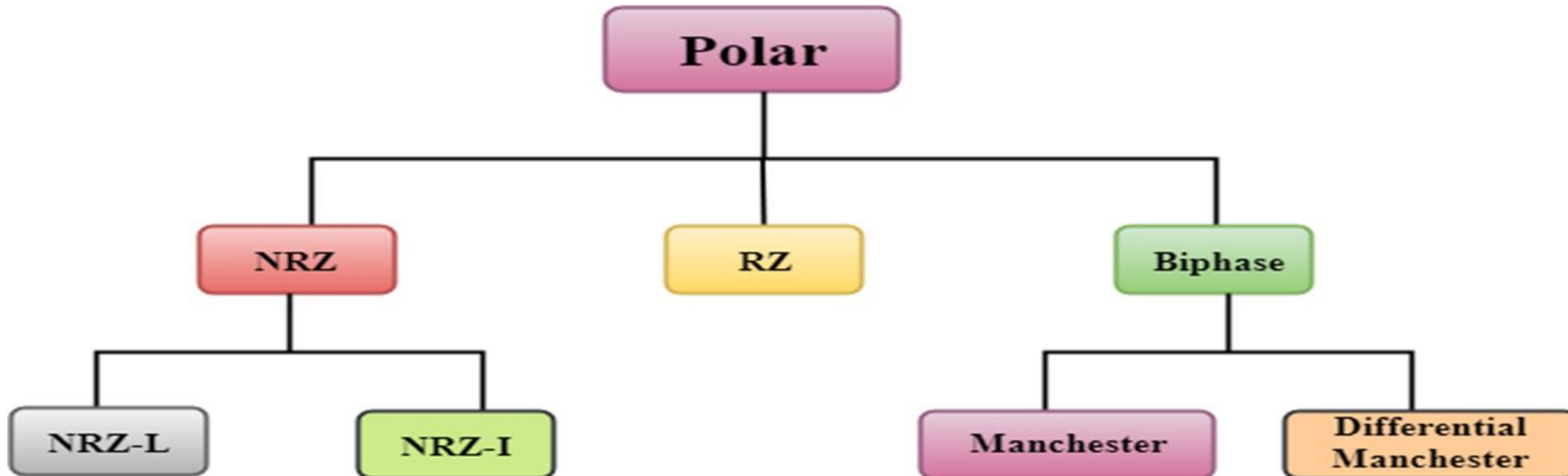
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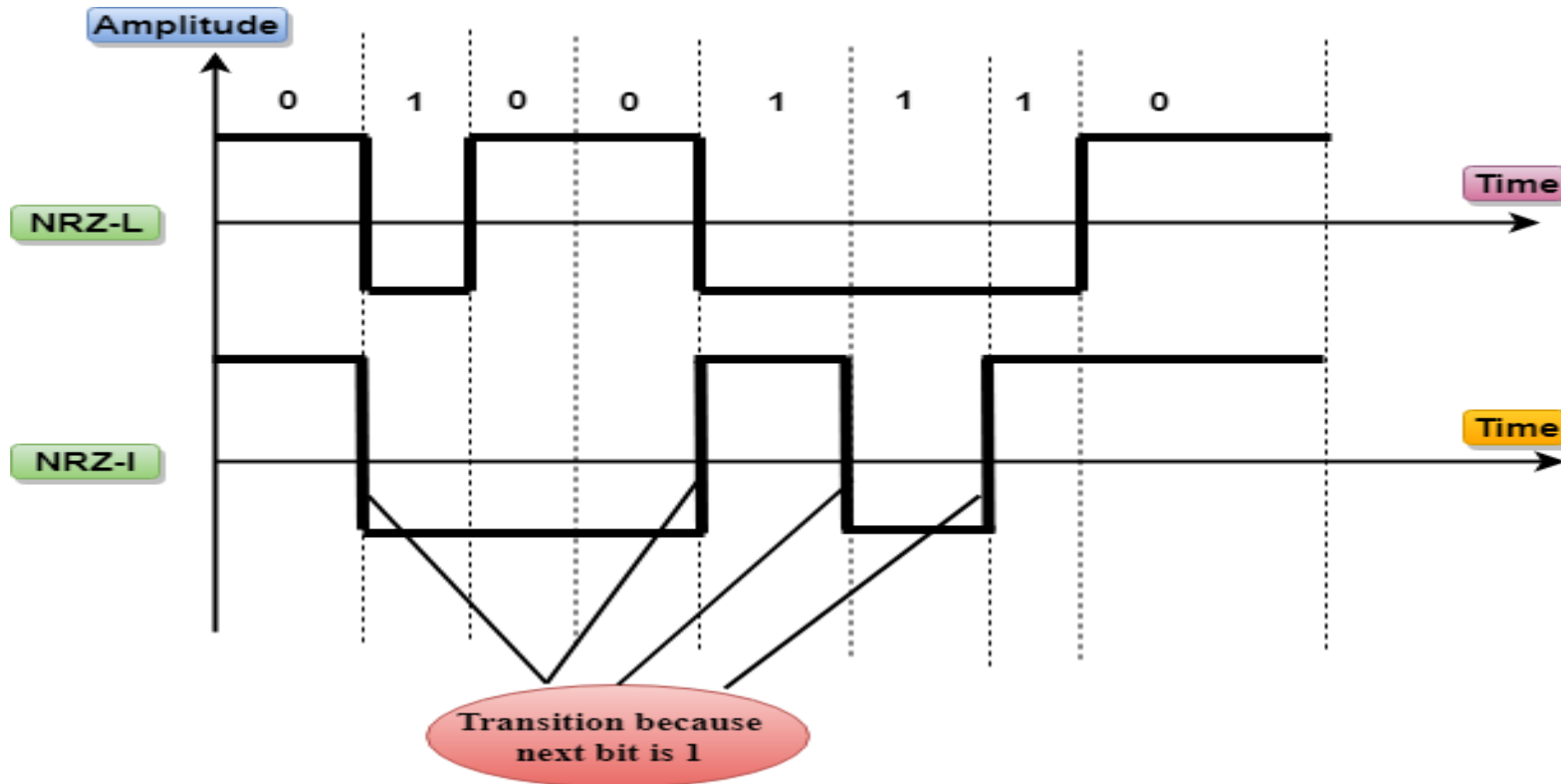
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Digital-to-digital Conversion

- NRZ



RZ(Return to zero)

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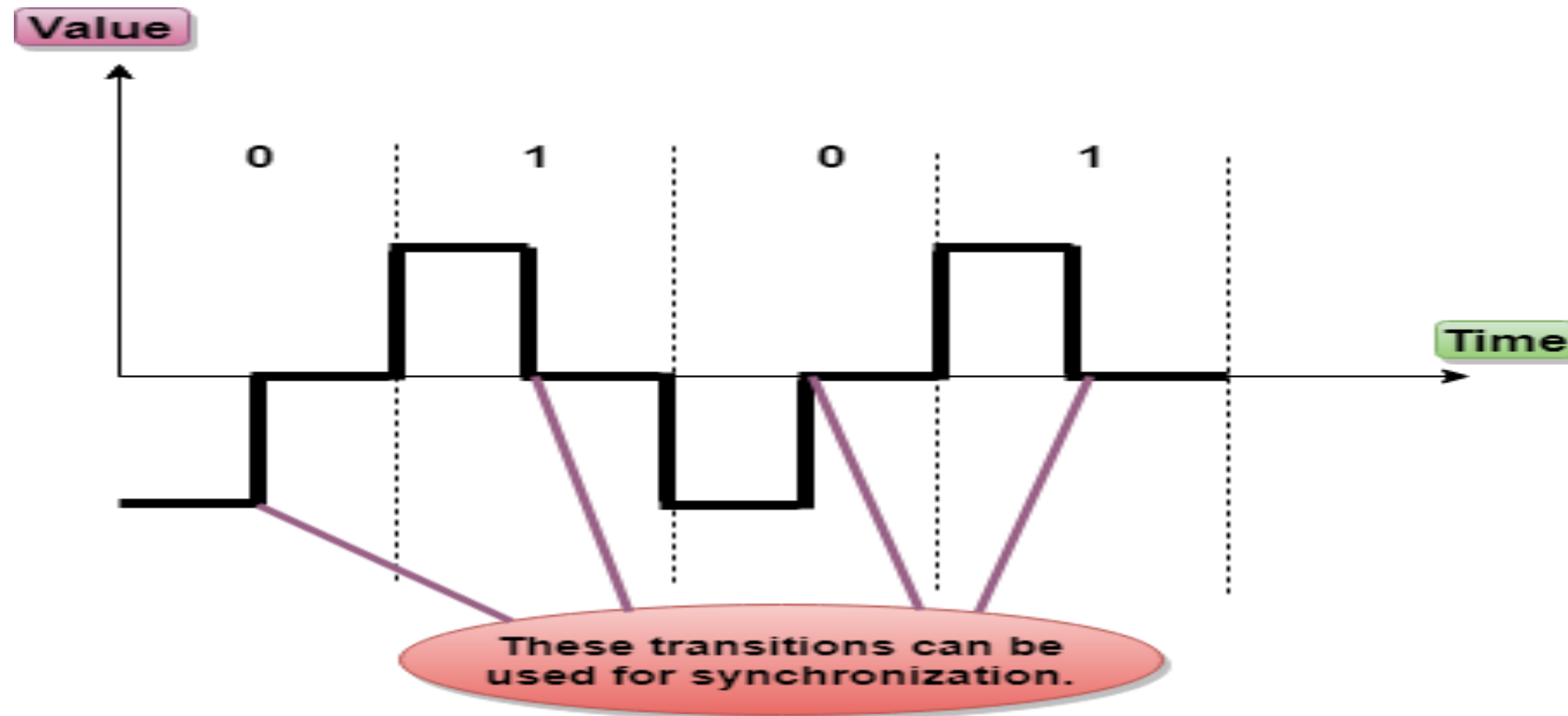
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Digital-to-digital Conversion

Disadvantage of RZ:

- It performs two signal changes to encode one bit that acquires more bandwidth.



Biphase

- Biphase is an encoding scheme in which signal changes at the middle of the bit interval but does not return to zero.
- Biphase encoding is implemented in two different ways –

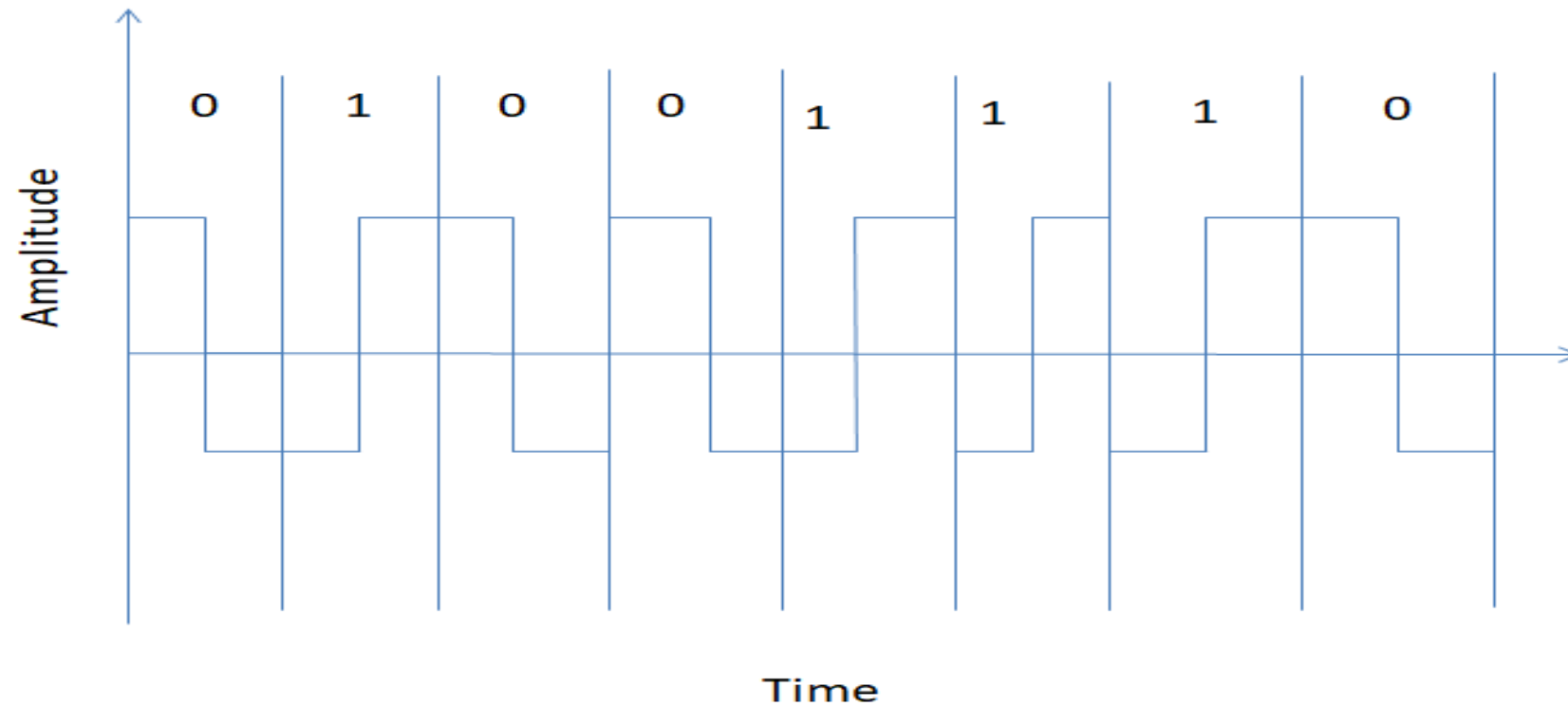
Manchester encoding scheme

Differential Manchester encoding scheme

Biphase – Manchester encoding scheme

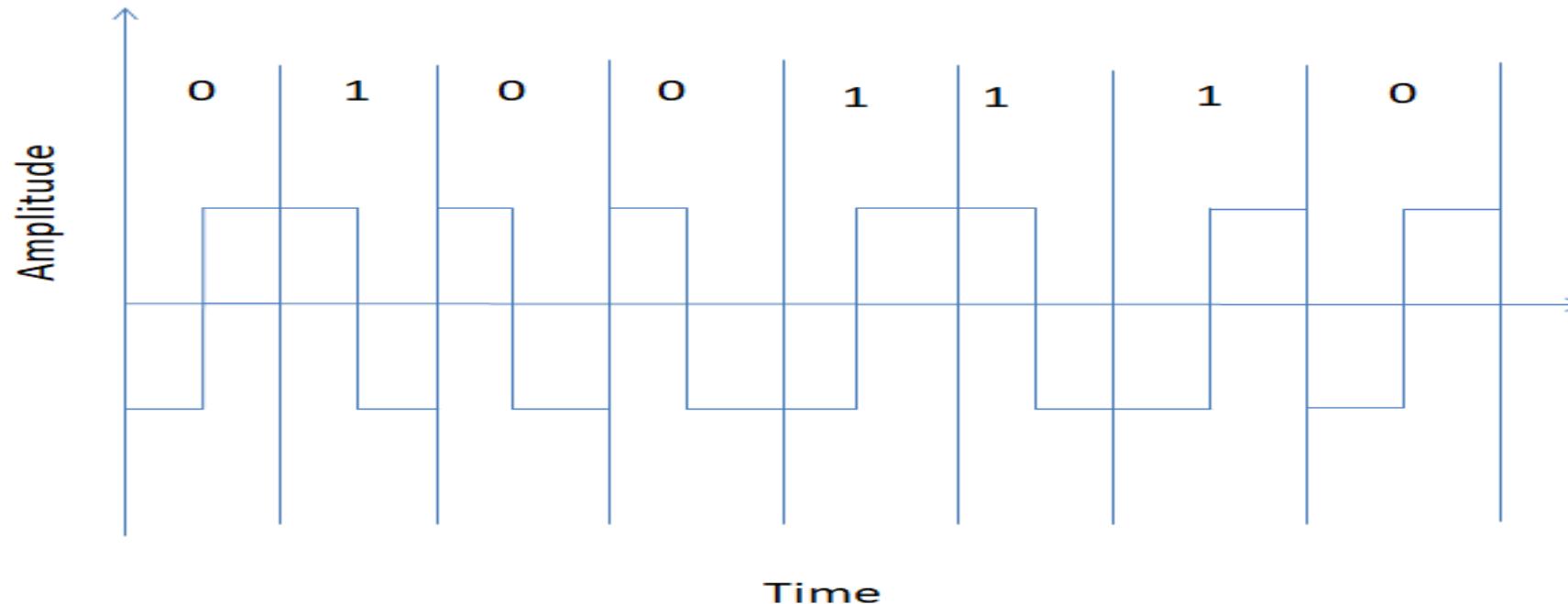
- It changes the signal at the middle of the bit interval but does not return to zero for synchronization.
- In Manchester encoding, a negative-to-positive transition represents binary 1 and positive-to-negative transition represents 0.
- Manchester has the same level of synchronization as RZ scheme except that it has two levels of amplitude.

Biphase – Manchester encoding scheme



Biphase – Differential Manchester encoding scheme

- The inversion at the middle of the bit is used. Transition is represented by binary 0 and no transition is represented by binary 1.





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Analog-to-Digital Conversion

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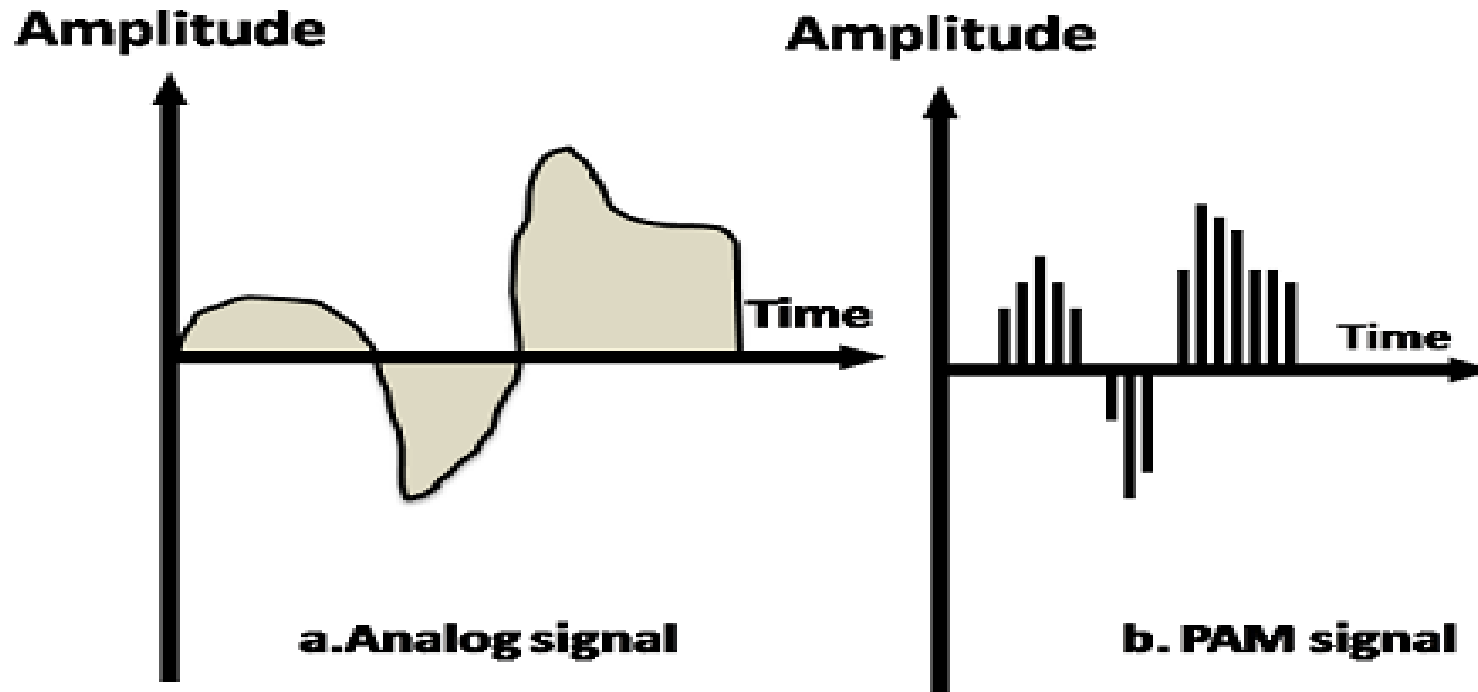
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- When an analog signal is digitalized, it is called an analog-to-digital conversion.
- Suppose human sends a voice in the form of an analog signal, we need to digitalize the analog signal. It requires a reduction in the number of values in an analog message so that they can be represented in the digital stream.
- In analog-to-digital conversion, the information contained in a continuous wave form is converted in **digital pulses**.

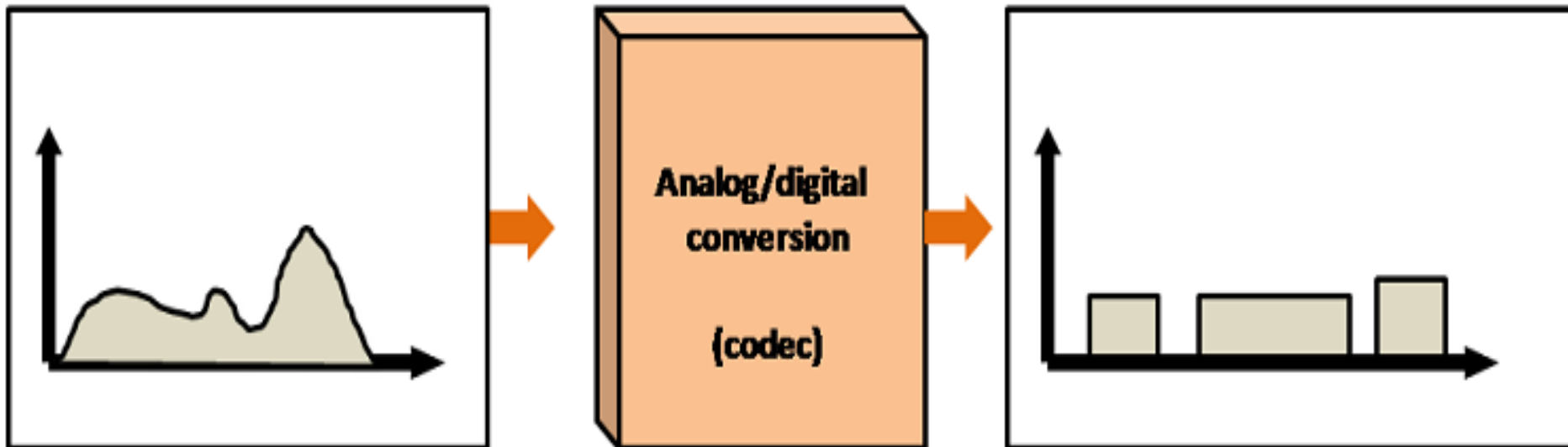
PAM - pulse amplitude modulation.

- PAM is a technique used in analog-to-digital conversion.
- PAM technique takes an analog signal, **samples it and generates a series of digital pulses based on the result of sampling.**
- Sampling means measuring the amplitude of a signal at equal intervals.
- PAM technique is not useful in data communication as it translates the original wave form into pulses, but these pulses are not digital. To make them digital, PAM technique is modified to PCM technique.

PAM – Pulse Amplitude Modulation



PAM – Pulse Amplitude Modulation



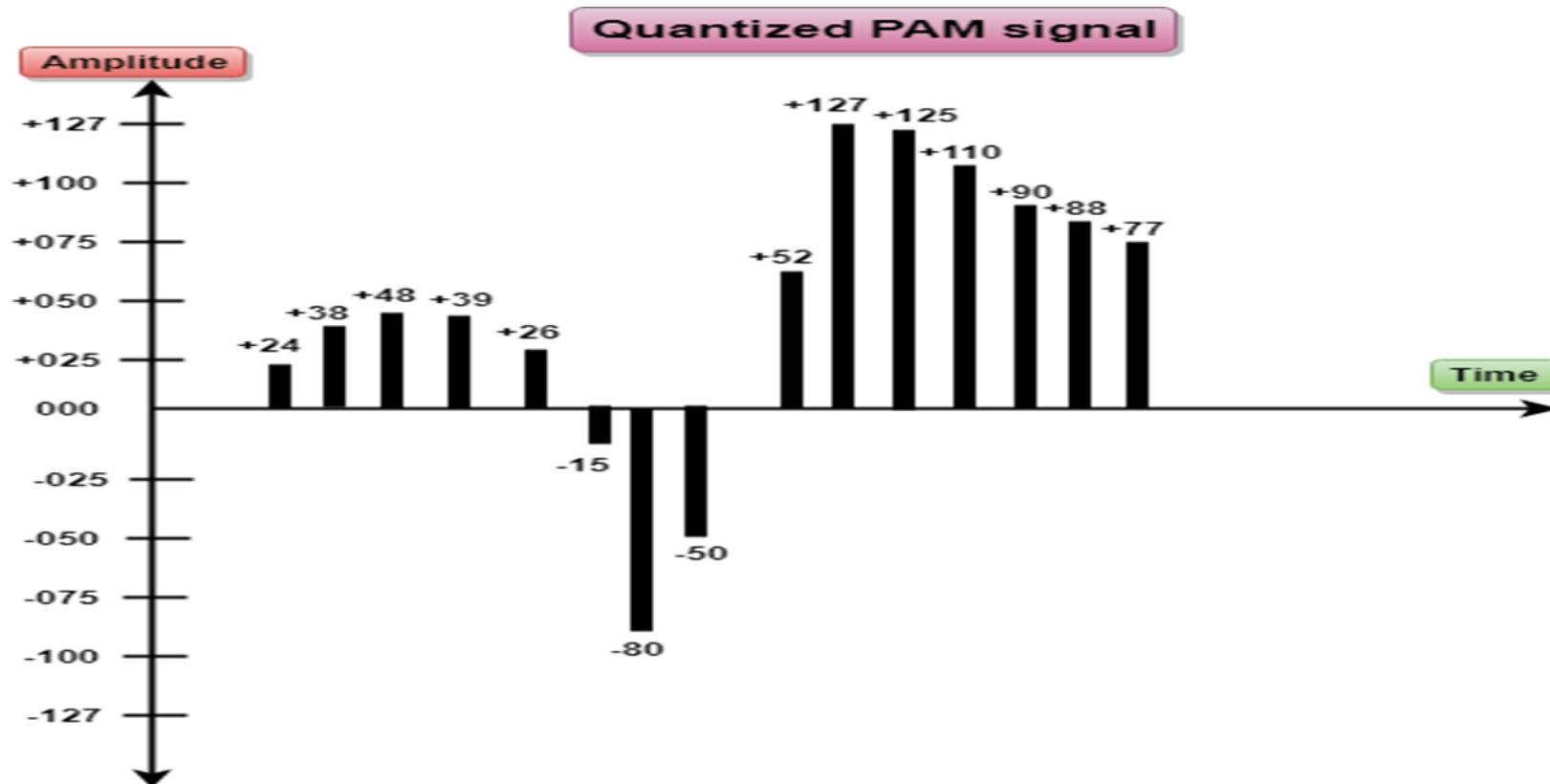
PCM – Pulse Code Modulation

- PCM technique is used to modify the pulses created by PAM to form a digital signal.
- PCM quantizes PAM pulses. **Quantization is a process of assigning integral values in a specific range to sampled instances.**
- PCM is made of four separate processes: **PAM, quantization, binary encoding, and digital-to-digital encoding.**

Data Communications

Techniques for Analog to-digital Conversion

PCM



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PCM





THANK YOU

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